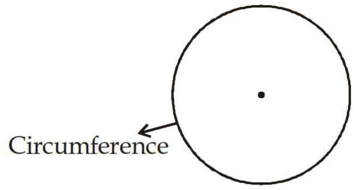
Chapter 16

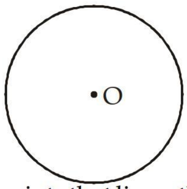
Circle

Circle : A circle is a set of points on a plane which lie at a fixed distance from a fixed point.

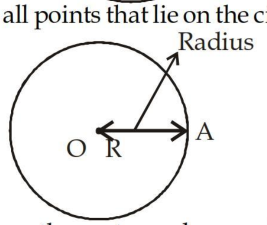
Circumference : The circumference of a circle is the distance around a circle which is equal to $2\pi r$. It is also called the perimeter of circle.



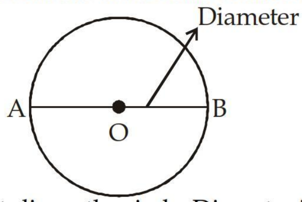
Centre : Fixed point is called the centre which is equidistant from all the points on the circumference. Here O is the center.



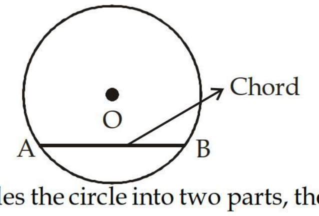
Radius : Fixed distance from the centre to all points that lie on the circumference.



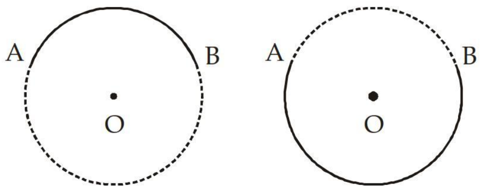
Diameter : A straight line which passes from the centre and connects two points of the circumference



Chord : A line segment whose end points lie on the circle. Diameter is also a largest chord.

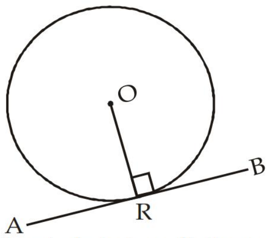


Arc : Any two points on the circle divides the circle into two parts, the smaller part is called as minor arc and the larger part is called as major arc.

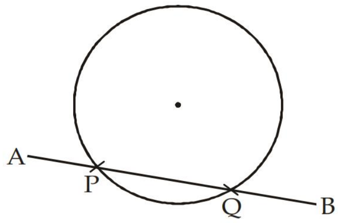


$\overset{\frown}{AB} \rightarrow$ Minor arc $\overset{\frown}{AB} \rightarrow$ Major arc

Tangent : A line segment which has one common point with the circumference of a circle i.e. it touches only at only one point is called as tangent of circle. AB $\rightarrow$ Tangent to circle at R.

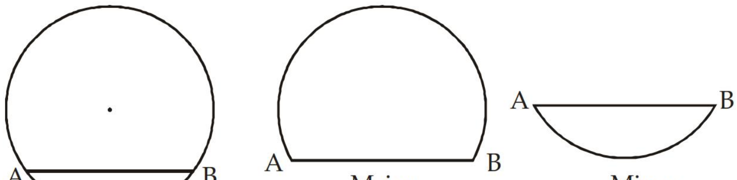


Secant : A line segment which intersects the circle in two distinct points, is called as secant. AB $\rightarrow$ Secant.



Segment : A chord divides a circle into two regions. These two regions are called the segments of a circle.

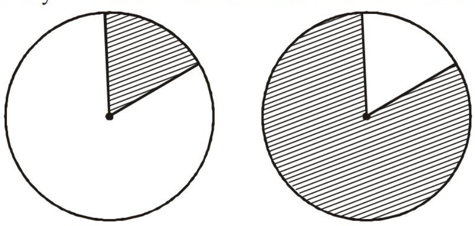
(a) major segment (b) minor segment



Major Segment

Minor Segment

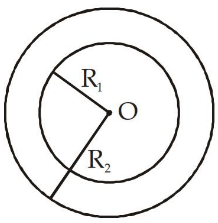
Sector : An area of circle enclosed by 2 radii and the circumference is called sector of circle.



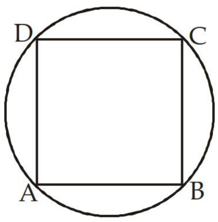
Minor Sector

Major Sector

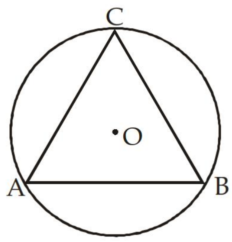
Concentric circles : Two circles having the same centre at a plane are called the concentric circles



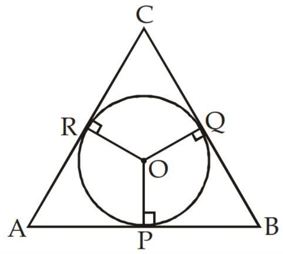
Cyclic Quadrilateral : A quadrilateral whose all the four vertices lie on the circle.

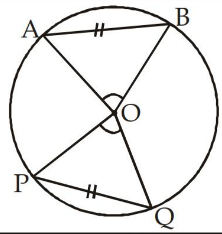
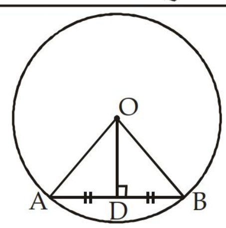
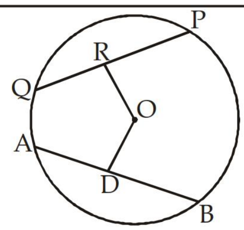
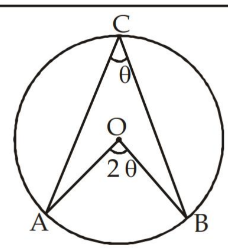


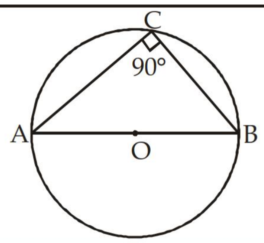
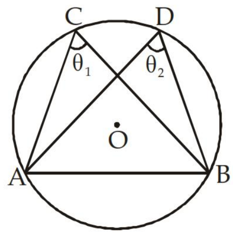
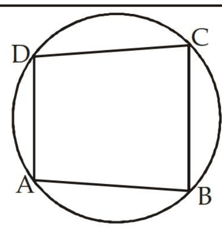
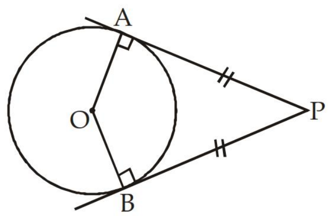
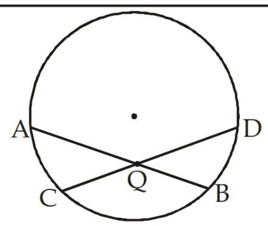
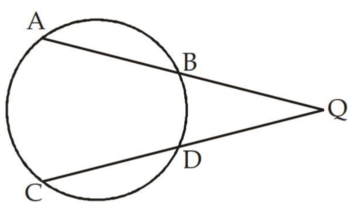
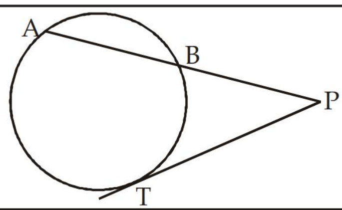
Circum-circle : A circle which passes through all the three vertices of a triangle.

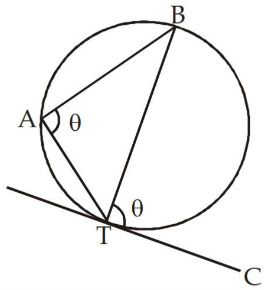
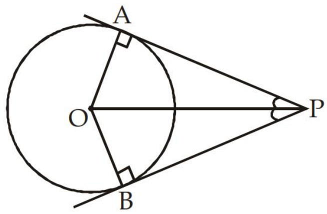
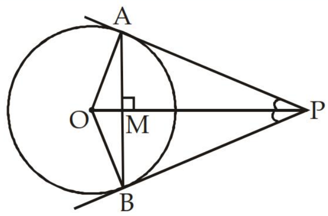


Incircle : A circle which touches all the three sides of a triangle i.e. all the three sides of a triangle are tangents to the circle is called an incircle



S. No.	Theorem	Diagram
1.	Equal Chords or Arc subtends equal angles at the centre $\widehat{PQ} = \widehat{AB}$ $\angle POQ = \angle AOB$	
2.	The perpendicular from the centre of a circle to a chord bisects the chord $OD \perp AB$ $AB = 2AD = 2BD$	
3.	Equal chords of circle are equidistant from the centre. $AB = PQ$ $OD = OR$	
4.	The angle subtended by an arc at the centre of a circle is twice the angle subtended by the arc at any point on remaining part of the circle $\angle AOB = 2m \angle ACB$	

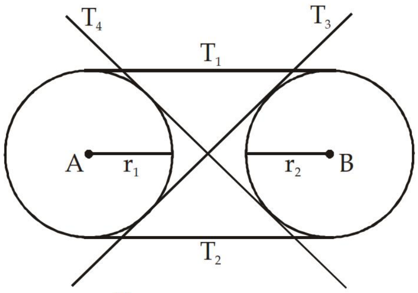
S. No.	Theorem	Diagram
5.	Angle in a semicircle is a right angle	
6.	Angles in the same segment of a circle are equal $\angle ACB = \angle ADB$ $\theta_1 = \theta_2$	
7.	The sum pair of opposite angles of a cyclic quadrilateral is 180°. $\angle DAB + \angle BCD = 180°$ $\angle ABC + \angle CDA = 180°$	
8.	The length of two tangents drawn from an external point to a circle are equal i.e. AP = BP	
9.	If two chords AB and CD of a circle, intersect inside a circle or outside a circle when when produced to at a point Q, then AQ × BQ = CQ × DQ.	
10.	When a chord AB is produced to meet a tangent PT at external point P then $PA \cdot PB = (PT)^2$	

S. No.	Theorem	Diagram
11.	**Alternate segment Theorem** → when a tangent is m drawn from point of contact of chord AB, then angle between chord and tangent will be equal to the Angle formed by the chord at the alternate segment. $\angle BTC = \angle TAB$	
12.	The Angle formed by two tangents meeting at on external point is bisected by a straight line. joining the centre of the circle to that external point $\angle BPO = \angle APO$ $\angle POB = POA$	
13.	When two tangents meet externally at point P and touch circle at A and B then PO is perpendicular bisector of AB $PO \perp AB$ and $AM = BM$	

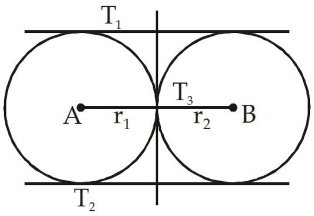
Number of common Tangents: The number of common tangent to the two circle are–

- Maximum $\longrightarrow$ 4
- Minimum $\longrightarrow$ Zero

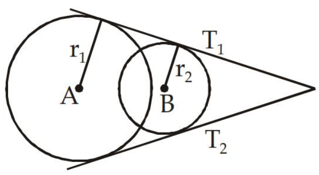
Case 1. 4 Common Tangents
Condition: $AB > r_1 + r_2$



Case 2. 3 Common Tangents
Condition: $AB = r_1 + r_2$

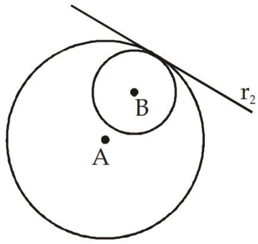


Case 3. 2 common Tangents
Condition $|r_1 - r_2| < AB < r_1 + r_2$



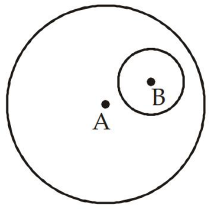
Case 4. 1 Common Tangent

Condition AB $\neq$ $|r_1 - r_2|$



Case 5. No Common Tangent

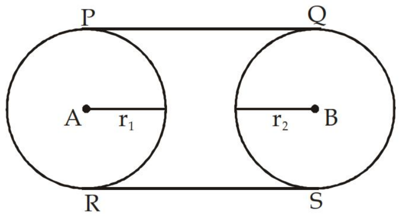
Condition AB < $|r_1 - r_2|$



There are two types of common tangent

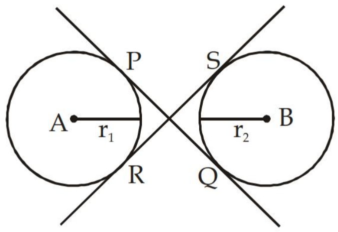
- ➲ Direct Common Tangent
- ➲ Transverse common tangent
- ➲ Length of Direct common Tangent (DCT)

$= \sqrt{(\text{Distance between centre})^2 - (r_1 - r_2)^2}$



- ➲ Length of Transverse common Tangent (TCT)

$= \sqrt{(\text{Distance between centre})^2 - (r_1 + r_2)^2}$



Foundation

Questions

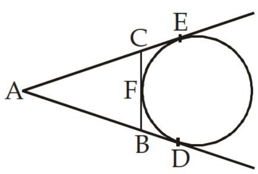
1. The distance between the centres of equal circles each of radius 3 cm is 10 cm. The length of a transverse common tangent is:

(a) 4 cm (b) 6 cm
(c) 8 cm (d) 10 cm

2. The number of common tangents that can be drawn to two given circles is at the most:

(a) one (b) two
(c) three (d) four

3. In the adjoining figure AD, AE and BC are tangents to the circle at D, E, F respectively



Then,

(a) AD = AB + BC + CA (b) 2AD = AB + BC + CA
(c) 3AD=AB + BC + CA (d) 4AD = AB + BC + CA

4. The distance between the centres of the two circles of radii r_1 and r_2 is d. They will touch each other internally if:

(a) $d = r_1$ or r_2 (b) $d = r_1 + r_2$
(c) $d = r_1 - r_2$ (d) $d = \sqrt{r_1 r_2}$

5. Through any given set of four points P, Q, R, S it is possible to draw:

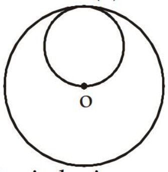
(a) atmost one circle (b) exactly one circle
(c) exactly two circles (d) exactly three circles

6. Two chords AB and CD of a circle intersect at E such that AE = 2.4 cm, BE = 3.2 cm and CE = 1.6 cm. The length of DE is:

(a) 1.6 cm (b) 3.2 cm
(c) 4.8 cm (d) 6.4 cm

7. In the adjoining figure, a smaller circle touches a larger circle internally and passes through the centre O of the larger circle. If the area of the smaller circle is 200 cm^2, the area of the larger circle in sq. cm is:

(a) 400 (b) 600
(c) 800 (d) 1000



8. The diameter of a circle circumscribing a square is 10 cm. Its side will be:

(a) 5 cm (b) $5\sqrt{2}$ cm
(c) 10 cm (d) $10\sqrt{2}$ cm

9. Three points A, B, C are on the same line. A circle passes through B and C. Then the focus of the tangent drawn from A to the circle, if the diameter of the circle is 2a, is:

(a) $x^2 + y^2 = a^2$ (b) $xx_1 + yy_1 = a^2$
(c) $xy = 0$ (d) $x + y = 0$

10. The chords which are equidistant from the centre of a circle are:

(a) equal (b) parallel
(c) perpendicular (d) None of these

11. The chord AB of a circle of centre O subtends an angle θ with the tangent at A to the circle. $\angle ABO$ is:

(a) θ (b) $2 \times (\pi - \theta)$
(c) $90° - \theta$ (d) $90° + \theta$

12. Any cyclic parallelogram having unequal adjacent sides is necessarily a:

(a) square (b) rectangle
(c) rhombus (d) trapezium

13. If a regular hexagon is inscribed in a circle of radius r, then its perimeter is:

(a) 6 r (b) 3 r
(c) 9 r (d) 12 r

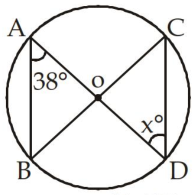
14. If a square ABCD is inscribed in a circle and AB = 4cm, then the radius of the circle is:

(a) 2 cm (b) $2\sqrt{2}$ cm
(c) 4 cm (d) $4\sqrt{2}$ cm

15. AB is the chord of circle and PQ is the tangent to circle at P. When AB is extented it meets Q. If BQ = 8, PQ = 12. Find the length of the chord.

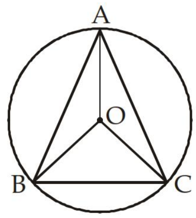
(a) 10 (b) 12
(c) 8 (d) 14

16. In the given figure O is the centre of the circle. If $\angle BAD$ = 38°, then $\angle OCD$ is:



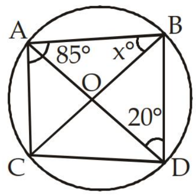
(a) 76° (b) 52°
(c) 38° (d) 19°

17. In the given figure, O is the centre of the circle. If $\angle OBC$ = 20°, the $\angle BAC$:



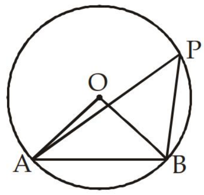
(a) 80° (b) 70°
(c) 100° (d) 140°

18. The value of x will be:



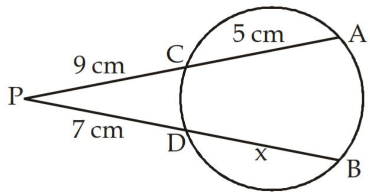
(a) 70° (b) 90°
(c) 60° (d) 75°

19. In the given figure, O is the centre of the circle and $\angle AOB$ = 90°, then $\angle APB$ will be:



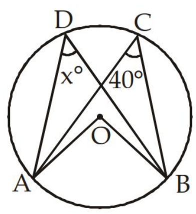
(a) 90° (b) 60°
(c) 45° (d) 30°

20. The value of x:



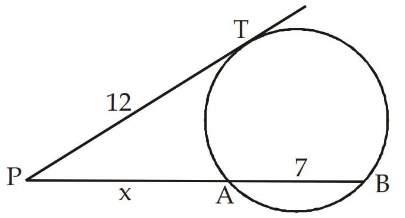
(a) 10 cm (b) 9 cm
(c) 7.5 cm (d) 11 cm

21. If O is centre of the circle, then x is equal to:



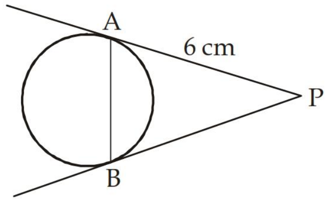
(a) 40° (b) 45°
(c) 39° (d) 35°

22. Find the value of x in the given figure:



(a) 16 cm (b) 9 cm
(c) 12 cm (d) 7 cm

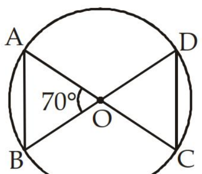
23. In the given figure, PA and PB are tangents from a point P to a circle such that PA = 6 cm and $\angle APB = 60°$. What is the length of the chord AB?



(a) 12 cm (b) 8 cm
(c) 9 cm (d) 6 cm

24. ABC is a right angled triangle AB = 3 cm, BC = 5 cm and AC = 4 cm, then the inradius of the circle is:
(a) 1 cm (b) 1.25 cm
(c) 1.5 cm (d) None of these

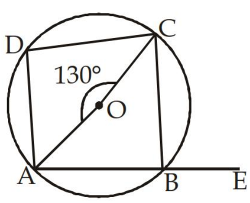
25. In the given figure, O is the centre of the circle. $\angle AOB = 70°$, find $\angle OCD$?



(a) 70° (b) 55°
(c) 65° (d) 110°

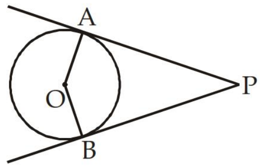
26. The distance between the centres of equal circles each of radius 3 cm is 10 cm. The length of a transverse tangent is:
(a) 4 cm (b) 6 cm
(c) 8 cm (d) 10 cm

27. In the given figure, $\angle AOC = 130°$. Find $\angle CBE$, where O is the centre?



(a) 130° (b) 100°
(c) 115° (d) 105°

28. In the given figure, O is the centre of the circle. PA and PB are tangents if $\angle AOB : \angle APB = 5 : 1$, then $\angle APB$?



(a) 150° (b) 30°
(c) 60° (d) 90°

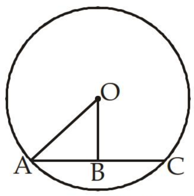
29. Two circles of radii 8 cm and 2 cm respectively touch each other externally at the point A. PQ is the direct common tangent of those two circles of centres O_1 and O_2 respectively. Then length of QP is equal to:
(a) 2 cm (b) 3 cm
(c) 4 cm (d) 8 cm

30. Two circles with radii 5 cm and 8 cm touch each other externally at a point A. If a straight line through the point A cuts the circles at points P and Q respectively, then AP : AQ is:
(a) 8 : 5 (b) 5 : 8
(c) 3 : 4 (d) 4 : 5

31. The radius of a circle is 6 cm. An external point is at a distance of 10 cm from the centre. Then the length of the tangent drawn to the circle from the external point upto the point of contact is:
(a) 8 cm (b) 10 cm
(c) 6 cm (d) 12 cm

32. The length of the chord of a circle is 8 cm and perpendicular distance between centre and the chord is 3 cm. Then the radius of the circle is equal to:
(a) 4 cm (b) 5 cm
(c) 6 cm (d) 8 cm

33. In the following figure, if OA = 10 and AC = 16, then OB must be, where OB is perpendicular to AC



(a) 5 (b) 6
(c) 3 (d) 4

34. The length of two chords AB and AC of a circle are 8 cm and 6 cm and $\angle BAC = 90°$, then the radius of circle is:
(a) 25 cm (b) 20 cm
(c) 30 cm (d) 5 cm

35. Circumcentre of ΔABC is O. If $\angle BAC = 85°$, $\angle BCA = 80°$, then $\angle AOC$ is:
(a) 80° (b) 30°
(c) 60° (d) 75°

36. If O is the circumcentre of ΔABC and $\angle OBC = 35°$, then $\angle BAC$ is equal to:
(a) 55° (b) 110°
(c) 70° (d) 35°

37. If S is the circumcentre of ΔABC and $\angle A = 50°$, then the value of $\angle BCS$ is:
(a) 20° (b) 40°
(c) 60° (d) 80°

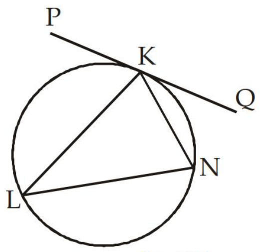
38. The distance between the centres of two equal circles, each of radius 4 cm, is 10 cm. The length of a transverse common tangent is:
(a) 8 cm (b) 10 cm
(c) 4 cm (d) 6 cm

39. A unique circle can always be drawn through x number of given non-collinear points, then x must be:
(a) 2 (b) 3
(c) 4 (d) 1

40. The length of radius of a circumcircle of a triangle having sides 3 cm, 4 cm and 5 cm is :
(a) 2 cm (b) 2.5 cm
(c) 3 cm (d) 1.5 cm

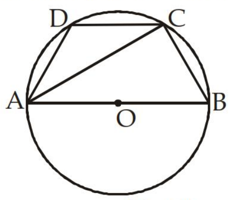
Moderate

1. In the given figure PKQ is a tangent and LN is the diameter of the circle. If $\angle KLN = 30°$ then $\angle PKL$ will be:



(a) 30° (b) 50°
(c) 60° (d) 70°

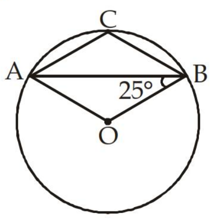
2. In the given figure $\angle ADC = 120°$ and AOB is the diameter of the circle, then $\angle BAC$:



(a) 30° (b) 40°
(c) 50° (d) 60°

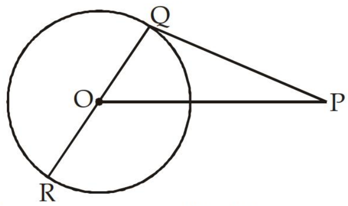
3. AB and CD are two parallel chords of a circle such that AB = 10 cm and CD = 24 cm, If the chords are on the opposite sides of the centre and the distance between them is 17 cm, then the radius of the circle is:
(a) 8 cm (b) 15 cm
(c) 11 cm (d) 13 cm

4. In the given figure, O is the centre of the circle then $\angle ACB$ will be:



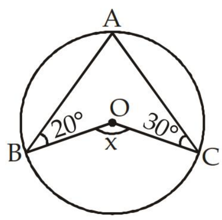
(a) 105° (b) 230°
(c) 115° (d) 100°

5. In the given figure, ROQ is the diameter of the circle. If $\angle POR = 120°$ then $\angle QPO$ will be:



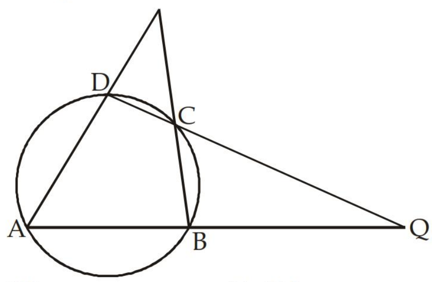
(a) 40° (b) 30°
(c) 60° (d) 50°

6. Find the value of $\angle x$ in the given figure:



(a) 120° (b) 130°
(c) 110° (d) 100°

7. In the adjoining figure $\angle A = 60°$ and $\angle ABC = 80°$, Find $\angle BQC$.



(a) 40° (b) 80°
(c) 20° (d) 30°

8. Two circles of radius 37 cm and 20 cm intersect each other at A and B. O and O' are the centres of the circles. If the length of AB is 24 cm, then OO':

(a) 50 cm (b) 51 cm
(c) 40 cm (d) 57 cm

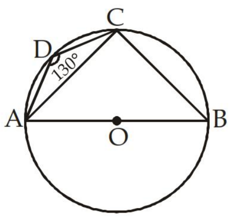
9. Two circles of radius 4 cm and 6 cm touch each other internally. Find the longest chord of the bigger circle which is out side the smaller circle?

(a) $8\sqrt{2}$ cm (b) $4\sqrt{2}$ cm
(c) $6\sqrt{2}$ cm (d) $3\sqrt{2}$ cm

10. AB is a chord of the circle (centre O). P is a point on the circle such that OP $\perp$ AB and OP intersect AB at point M. If AB = 8 cm, MP = 2 cm then radius (r):

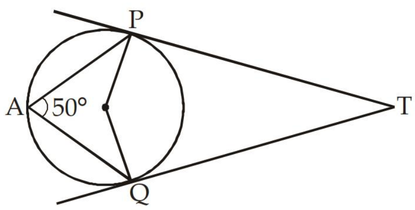
(a) 7 cm (b) 5 cm
(c) 6 cm (d) 4 cm

11. In the given figure, ABCD is a cyclic quadrilateral whose side AB is a diameter of the circle. If $\angle$ADC = 130° find $\angle$CAB.



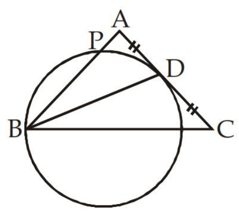
(a) 40° (b) 50°
(c) 30° (d) 130°

12. In the given figure, TP and TQ are tangents to the circle. If $\angle$PAQ = 50°, what is $\angle$PTQ?



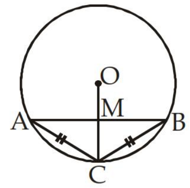
(a) 80° (b) 70°
(c) 100° (d) 90°

13. In the figure, ABC is a triangle in which AB = AC. A circle through B touches AC at D and intersects AB at P. If D is the mid-point of AC, Find the value of AB:



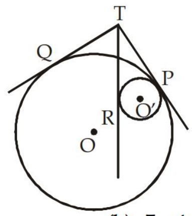
(a) 2AP (b) 3AP
(c) 4AP (d) None of these

14. In the given figure, the chords AC and BC are equal. The radius OC intersect AB at M then AM : BM:



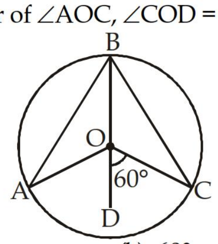
(a) 1 : 1 (b) $\sqrt{2}:3$
(c) $3:\sqrt{2}$ (d) None of these

15. In the given figure, Tangents TQ and TP are drawn to the larger circle with centre O and tangents TP and TR are drawn to the smaller circle (centre O'). Find TQ : TR.



(a) 1 : 1 (b) 5 : 4
(c) 8 : 7 (d) 7 : 8

16. 'O' is the centre of the circle, line segment BOD is the angle bisector of $\angle$AOC, $\angle$COD = 60°. Find $\angle$ABC:



(a) 120° (b) 60°
(c) 30° (d) 90°

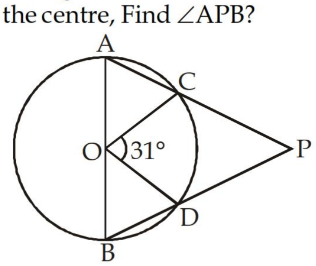
17. ABCD is a cylic quadrilateral. Side AB and DC when produced meet at P and side AD and BC when Produced meet at Q. If $\angle$APD = 40°, $\angle$ADC = 85°, then $\angle$AQB is equal to:

(a) 30° (b) 40°
(c) 50° (d) 55°

18. AB is the diameter of a circle whose center is O and CD is a chord in the circle and CD = $\frac{1}{2}$AB. AC and BD when produced meet at P. Find $\angle$APB?

(a) 30° (b) 40°
(c) 50° (d) 60°

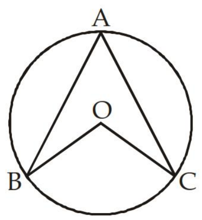
19. In the given figure, AB is the diameter of the circle and O is the centre, Find $\angle$APB?



(a) 149° (b) 74.5°
(c) 62° (d) None of these

20. A, B, C are three points on a circle. The tangent at A meets BC produced at T, $\angle BTA = 40°$ and $\angle CAT = 44°$ The angle subtended by BC at the centre of the circle is:
(a) 84° (b) 92°
(c) 96° (d) 104°

21. BC is the chord of a circle with centre O. A is a point on major arc BC as shown in the above figure. What is the value of $\angle BAC + \angle OBC$?



(a) 120° (b) 60°
(c) 90° (d) 180°

22. AB and CD are two parallel chords drawn on two opposite sides of the diameter such that AB = 6 cm, CD = 8 cm. If the radius of the circle is 5 cm, the distance between the chords, in cm, is:
(a) 2 (b) 7
(c) 5 (d) 3

23. Two tangents are drawn from a point P to a circle at A and B. O is the centre of the circle. If $\angle AOP = 60°$, then $\angle APB$ is:
(a) 120° (b) 90°
(c) 60° (d) 30°

24. If the length of a chord of a circle, which makes an angle 45° with the tangent drawn at one end point of the chord, is 6 cm, then the radius of the circle is:
(a) $6\sqrt{2}$ cm (b) 5 cm
(c) $3\sqrt{2}$ cm (d) 6 cm

25. The radius of two concentric circles are 9 cm and 15 cm. If the chord of the greater circle be a tangent to the smaller circle, then the length of that chord is:
(a) 24 cm (b) 12 cm
(c) 30 cm (d) 18 cm

26. AB = 8 cm and CD = 6 cm are two parallel chords on the same side of the centre of a circle. The distance between them is 1 cm. The radius of the circle is:
(a) 5 cm (b) 4 cm
(c) 3 cm (d) 2 cm

27. What is the ratio of Inradius and circumradius of right angle triangle?
(a) 1 : 1 (b) 2 : 1
(c) 1 : 2 (d) None of these

28. Two circles touch each other externally at point A and PQ is a direct common tangent which touches circle at P and Q respectively. Then $\angle PAQ$ =
(a) 45° (b) 90°
(c) 80° (d) 100°

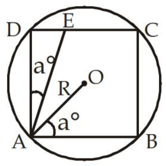
29. AB is a chord to a circle and PAT is the tangent to the circle at A. If $\angle BAT = 75°$ and $\angle BAC = 45°$, C being a point on the circle, then $\angle ABC$ is equal to:
(a) 40° (b) 45°
(c) 60° (d) 70°

30. The circumcentre of a triangle ABC is O. If $\angle BAC = 85°$ and $\angle BCA = 75°$, then the value of $\angle OAC$ is:
(a) 40° (b) 60°
(c) 70° (d) 90°

Difficult

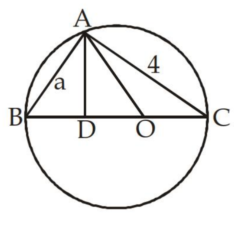
1. AB and AC are two chords of a circle such that AB = AC = 6 cm. If radius of the circle is 5 cm, then BC is:
(a) 4.8 cm (b) 9.6 cm
(c) 2.4 cm (d) 8.4 cm

2. From the figure given below find AE : AD where Area (circle) : Area (Square) = $\pi : \sqrt{3}$ O is the centre of circle and ABCD is the square inscribed in circle. R is the radius of circle



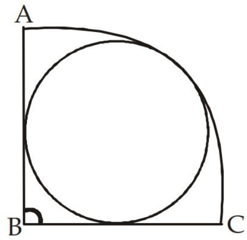
(a) $1 : \sqrt{2}$ (b) $\sqrt{2} : 1$
(c) $2 : \sqrt{3}$ (d) $\sqrt{3} : 2$

3. If in the given figure, AB = a, AC = 4 cm, while O is the centre of the circle and D is a point between O and B such that AD ⊥ BC. Find the length of OD?



(a) $\frac{4-a}{4}$ (b) $\frac{16-a^2}{2\sqrt{a^2+16}}$
(c) $\frac{4a-16}{16a-a^2}$ (d) $\frac{2\sqrt{a^2-16}}{16+a^2}$

4. If ABC is a Quarter Circle and a circle is inscribed in it and if AB = 1 cm, find radius of smaller circle.



(a) $\sqrt{2}-1$ (b) $\frac{\sqrt{2}-1}{2}$

(c) $\frac{\sqrt{2}+1}{2}$ (d) $1-2\sqrt{2}$

5. AB and CD are two chords of a circle which intersect at right angle at E. If AE = 2 cm, EB = 6 cm, ED = 3 cm, then radius (r) is equal to:

(a) $\frac{\sqrt{65}}{2}$ (b) $\sqrt{65}$

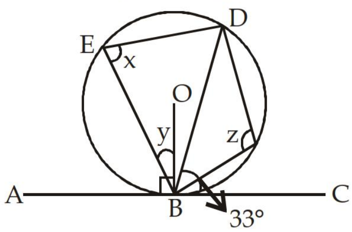
(c) $2\sqrt{65}$ (d) None of these

6. Three equal circle of unit radius touch each other. Then, the area of the circle circumscribing the three circle is:

(a) $6\pi\left(2+\sqrt{3}\right)^2$ (b) $\frac{\pi}{6}\left(2+\sqrt{3}\right)^2$

(c) $\frac{\pi}{3}\left(2+\sqrt{3}\right)^2$ (d) $3\pi\left(2+\sqrt{3}\right)^2$

7. In the given figure, chord BE = BD, $\angle CBD = 33°$, and OB $\perp$ AC then x + y + z is equal to:



(a) 230° (b) 237°
(c) 337° (d) None of these

8. Two chords AB and CD of circle whose centre is O, meet at the point P and $\angle AOC = 50°$, $\angle BOD = 40°$. Then the value of $\angle BPD$ is:
(a) 60° (b) 40°
(c) 45° (d) 75°

9. A circle (with centre at O) is touching two intersecting lines AX and BY. The two points of contact A and B subtend an angle of 65° at any point C on the circumference of the circle. If P is the point of intersection of the two lines, then the measure of $\angle APO$ is:
(a) 25° (b) 65°
(c) 90° (d) 40°

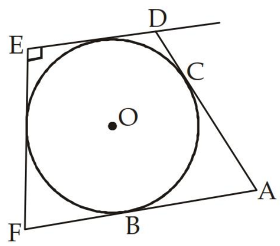
10. The tangents are drawn at the extremities of a diameter AB of a circle with centre P. If a tangent to the circle at the point C intersects the other two tangents at Q and R, then the measure of the $\angle QPR$ is:
(a) 45° (b) 60°
(c) 90° (d) 180°

11. In a ΔABC, I and O are the in-centre and circum-centre respectively. The line AI is produced to a point D on the circumcircle. If $\angle BOD = Z$, $\angle BID = Y$ and $\angle ABC = X$, then $\frac{x+z}{3y}$ is equal to:

(a) $\frac{2}{3}$ (b) $\frac{1}{3}$

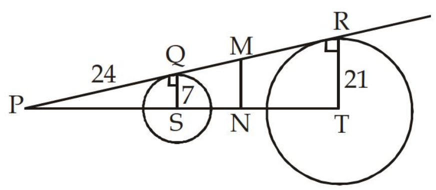
(c) $\frac{4}{3}$ (d) 1

12. In the given figure, AB = 27, AD = 38 cm, ED = 24 cm and $\angle E = 90°$, then radius of the circle is equal to:



(a) 11 cm (b) 15 cm
(c) 13 cm (d) 17 cm

13. In the given figure, PQ = 24 cm. M is the mid-point of QR?



Also, MN $\perp$ PR, QS = 7 cm and TR = 21 cm, then SN = ?
(a) 50 cm (b) 12.5 cm
(c) 31 cm (d) 25 cm

14. Two circles having radius 'a' cm and 'b' cm touch each other externally. Another circle whose radius is 'c' cm, touches both the circles and also their common tangent. Then which statement will be true?

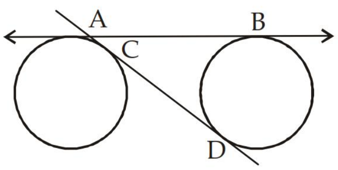
(a) $\sqrt{a}+\sqrt{b}=\sqrt{c}$ (b) $\sqrt{a}=\sqrt{b}+\sqrt{c}$

(c) $\sqrt{ab}+\sqrt{bc}=\sqrt{ac}$ (d) $\frac{1}{\sqrt{a}}+\frac{1}{\sqrt{b}}=\frac{1}{\sqrt{c}}$

15. ABC and MNC are two secants of a circle whose centre is O. AN is the diameter of the circle if $\angle BAN = 38°$ and $\angle ACM = 20°$ then $\angle MBN$:
(a) 38° (b) 42°
(c) 28° (d) 32°

16. Two tangents PA and PB are drawn to the circle (centre O) from a point P. CD is another tangent on the circle which intersects PA and PB at C and D respectively. If $\angle APB = 34°$ then $\angle COD$?
(a) 146° (b) 68°
(c) 73° (d) None of these

17. If two equal circles of radius 5 cm have two common tangent AB and CD which touch the circle on A, C, and B, D respectively and if CD = 24 cm, find the length of AB?



(a) 27 cm (b) 25 cm
(c) 26 cm (d) 30 cm

18. '2a' and '2b' are the length of two chords which intersect at right angle. If the distance between the centre of the circle and the intersecting point of the chords is 'C' then the radius of the circle is:
(a) $\frac{\sqrt{a^2b^2c^2}}{2}$ (b) $\sqrt{a^2+b^2+c^2}$
(c) $\sqrt{\frac{a^2+b^2+c^2}{2}}$ (d) None of these

19. P and Q are the mid points of two chords (not diameters) AB and AC respectively of a circle with centre at a point O. The lines OP and OQ are produced to meet the circle respectively at the points R and S. T is any point on the major arc between the points R and S of the circle. If $\angle BAC = 32°$, $\angle RTS = ?$
(a) 32° (b) 74°
(c) 106° (d) 64°

20. Two circles of radii 9 cm and 2 cm respectively have centres X and Y and $\overline{XY} = 17$ cm. Circle of radius r cm with centre Z touches two given circles externally. If $\angle XZY = 90°$, find r:
(a) 18 cm (b) 3 cm
(c) 12 cm (d) 6 cm

Previous Year (Memory Based)

1. Two chords of lengths a m and b m subtend angles 60° and 90° at the centre of the circle, respectively. Which of the following is true?
(a) $b = \sqrt{2}a$ (b) $a = \sqrt{2}b$
(c) a = 2b (d) b = 2a

2. The radius of a circle is 6 cm. The distance of a point lying outside the circle from the centre is 10 cm. The length of the tangent drawn from the outside point to the circle is:
(a) 5 cm (b) 6 cm
(c) 7 cm (d) 8 cm

3. Two chords AB and CD of a circle with centre O, intersect each other at P. If $\angle AOD = 100°$ and $\angle BOC = 70°$, then the value of $\angle APC$ is:
(a) 80° (b) 75°
(c) 85° (d) 95°

4. Radii of two circles are 6.3 cm and 3.6 cm. If they touch each other internally, then the distance between their centres is:
(a) 9.1 cm (b) 2.7 cm
(c) 3.7 cm (d) 10.1 cm

5. There is a circle of radius 5 cm and the perpendicular distance from the centre of the circle to the chord of the circle is 3 cm. Then, the length of the chord is:
(a) 6 cm (b) 5 cm
(c) 8 cm (d) 4 cm

6. The bisector of $\angle BAC$ of ΔABC cuts BC at D and the circumcircle of the triangle at E. If DE = 3 cm, AC = 4 cm and AD = 5 cm, then the length of AB is:
(a) 9 cm (b) 10 cm
(c) 7 cm (d) 8 cm

7. If an equilateral ΔABC is inscribed in a circle with centre at O, then $\angle BOC$, $\angle COA$ and $\angle AOB$ are respectively:
(a) 50°, 60°, 70° (b) 120°, 120°, 120°
(c) 60°, 60°, 60° (d) 80°, 120°, 160°

8. Two circles intersect each other at the points A and B. A straight line parallel to AB intersects the circles at C, D, E and F. If CD = 4.5 cm, then the measure of EF is:
(a) 1.50 cm (b) 2.25 cm
(c) 4.50 cm (d) 9.00 cm

9. Two circles with radii 25 cm and 9 cm touch each other externally. The length of the direct common tangent is:
(a) 34 cm (b) 30 cm
(c) 36 cm (d) 32 cm

10. Two circles touch each other externally at a point. Then, the number of direct common tangents the circles may have, is:
(a) 2 (b) 4
(c) 3 (d) 1

11. Find the distance of the chord of length 16 cm from the centre of the circle of diameter 20 cm?
(a) 6 cm (b) 8 cm
(c) 12 cm (d) 16 cm

12. If the circumcentre of a triangle lie on the side whose adjacent angles are 45° each, then find the other two sides, if the radius of the circumcircle is 15 cm?
(a) 15 cm (b) 30 cm
(c) $15\sqrt{2}$ cm (d) $30\sqrt{2}$ cm

13. Let C be a point on a straight line AB. Circles are drawn with diameters AC and AB. Let P be any point on the circumference of the circle with diameter AB. If AP meets the other circle at Q, then
(a) QC | | PB
(b) QC is never parallel of PB
(c) $QC = \frac{1}{2} PB$
(d) QC | | PB and $QC = \frac{1}{2} PB$

14. If O be the circumcentre of a ΔPQR and $\angle QOR = 110°$, $\angle OPR = 25°$, then the measure of $\angle PRQ$ is:
(a) 50° (b) 55°
(c) 60° (d) 65°

15. In ΔABC, O is its circumcentre and $\angle BAC = 50°$. The measure of $\angle OBC$ is:
(a) 60° (b) 30°
(c) 40° (d) 50°

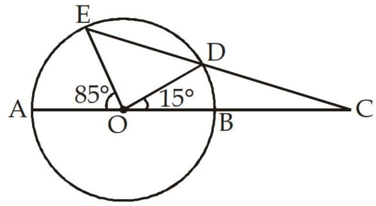
16. Two circles of same radius 5 cm, intersect each other at A and B. If AB = 8 cm, then the distance between the centres is:
(a) 10 cm (b) 4 cm
(c) 6 cm (d) 8 cm

17. N is the foot of the perpendicular from a point P of a circle with radius 7 cm, on a diameter AB of the circle. If the length of the chord PB is 12 cm, the distance of the point N from the point B is:
(a) $12\frac{2}{7}$ cm (b) $3\frac{5}{7}$ cm
(c) $10\frac{2}{7}$ cm (d) $6\frac{5}{7}$ cm

18. A circle is touching the side BC of ΔABC at P and is also touching AB and AC produced at Q and R, respectively. If AQ = 6 cm, then perimeter of ΔABC is:
(a) 6 cm (b) 10 cm
(c) 12 cm (d) 18 cm

19. The length of two parallel chords of a circle are 6 cm and 8 cm. If the smaller chord is at a distance of 4 cm from the centre, the distance of the other chord from the centre is: (chords are on opposite side of centre)
(a) 5 cm (b) 6 cm
(c) 4 cm (d) 3 cm

20. In the adjoining figure, AB is a diameter of the circle with centre O. If $\angle BOD = 15°$ and $\angle EOA = 85°$, the measure of $\angle ECA$ is:



(a) 45° (b) 35°
(c) 30° (d) 70°

21. If the incentre of an equilateral triangle lies inside the triangle and its inradius is 3 cm, the side of the equilateral triangle is:
(a) $9\sqrt{3}$ cm (b) $6\sqrt{3}$ cm
(c) $3\sqrt{3}$ cm (d) 6 cm

22. The circumcentre of a ΔABC is O. If $\angle BAC = 85°$ and $\angle BCA = 75°$, then the value of $\angle OAC$ is:
(a) 40° (b) 60°
(c) 70° (d) 90°

23. Two parallel chords are drawn in a circle of diameter 30 cm. The length of one chord is 24 cm and the distance between the two chords is 21 cm. The length of the other chord is:
(a) 10 cm (b) 18 cm
(c) 12 cm (d) 16 cm

24. If two equal circles, whose centres are O and O', intersect each other at the points A and B, OO' = 12 cm and AB = 16 cm, then the radius of the circles is:
(a) 10 cm (b) 8 cm
(c) 12 cm (d) 14 cm

25. A chord of a circle is equal to its radius. The angle subtended by this chord at a point on the circumference in the major segment is:
(a) 60° (b) 120°
(c) 90° (d) 30°

26. Two circles of radii 9 cm and 2 cm, respectively has centres X and Y and $\overline{XY} = 17$ cm. Circle of radius r cm with centre Z touches two given circles externally. If $\angle XZY = 90°$, find r?

(a) 18 cm (b) 3 cm
(c) 12 cm (d) 6 cm

27. A circle (with centre at O) is touching two intersecting lines AX and BY. The two points of contact A and B subtend an angle of 35° at any point C on the circumference of the circle. If P is the point of intersection of the two lines, then the measure of ∠APO is:
(a) 55° (b) 65°
(c) 90° (d) 40°

28. Two chords AB and CD of a circle with centre O intersect each other at the point P. If ∠AOD = 20° and ∠BOC = 30°, then ∠BPC is equal to:
(a) 50° (b) 20°
(c) 25° (d) 30°

29. The distance between the centres of two equal circles each of radius 3 cm is 10 cm. The length of a transverse common tangent is:
(a) 4 cm (b) 6 cm
(c) 8 cm (d) 10 cm

30. The length of the common chords of two intersecting circles is 24 cm. If the diameters of the circles are 30 cm and 26 cm, then the distance between the centres in cm is:
(a) 13 (b) 14
(c) 15 (d) 16

31. AC is the diameter of a circumcircle of ΔABC, Chord ED is parallel to the diameter AC. If ∠CBE = 50°, then the measure of ∠DEC is:
(a) 50° (b) 90°
(c) 60° (d) 40°

32. A chord of length 8 cm is at a distance of 3 cm from the centre of the circle. The length of the radius of the circle is:
(a) $\sqrt{73}$ cm (b) $\sqrt{55}$ cm
(c) 5 cm (d) 10 cm

33. From a point P, two tangents PA and PB are drawn to a circle with centre O. If OP is equal to diameter of the circle, then ∠APB is:
(a) 90° (b) 30°
(c) 60° (d) 45°

34. A chord 12 cm long is drawn in a circle of diameter 20 cm. The distance of the chord from the centre is:
(a) 6 cm (b) 10 cm
(c) 16 cm (d) 8 cm

35. P and Q are two points on a circle with centre at O. R is a point on the minor arc of the circle, between the points P and Q. The tangents to the circle at the points P and Q meet each other at the point S. If ∠PSQ = 20°, then ∠PRQ = ?
(a) 80° (b) 200°
(c) 160° (d) 100°

36. The radius of the circumcircle of a right angled triangle is 15 cm and the radius of its in-circle is 6 cm. Find the sides of the triangle?
(a) 30, 40, 41 (b) 18, 24, 30
(c) 30, 24, 25 (d) 24, 36, 20

37. The radius of two concentric circles are 17 cm and 10 cm. A straight line ABCD intersects the larger circle at the point A and D and intersects the smaller circle at the points B and C. If BC = 12 cm, then the length of AD (in cm) is:
(a) 20 (b) 24
(c) 30 (d) 34

38. If the radii of two circles be 6 cm and 3 cm and the length of transverse common tangent be 8 cm, then the distance between the two centres is
(a) $\sqrt{145}$ cm (b) $\sqrt{140}$ cm
(c) $\sqrt{150}$ cm (d) $\sqrt{135}$ cm

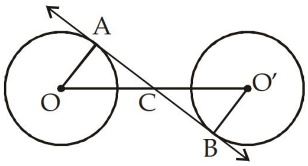
39. The exterior angles obtained on producing the base BC of a triangle ABC in both ways are 120° and 105°, then the vertical ∠A of the triangle is?
(a) 36° (b) 40°
(c) 45° (d) 55°

40. Two circles touch each other externally at point P. QR is a direct common tangent at Q, R respectively. NP is the direct common tangent at P where N lies on QR. Find ∠QPR
(a) 30° (b) 60°
(c) 90° (d) 120°

Foundation

Solutions

1. **(c);** The transverse common tangent to two equal circles bisects the line joining their centres.



OO' = 10 cm

OA = O'B = 3 cm

$$AB = \sqrt{10^2 - 6^2} = 8 \text{ cm}$$

2. **(d);** At most four common tangents can be drawn to two circles.

3. **(b);** BD = BF (tangent through a common point)

CE = CF (tangent through a common point)

AD = AE (tangent through a common point)

AD = AB + BD = AB + BF ...(i)

AE = AC + CF ...(ii)

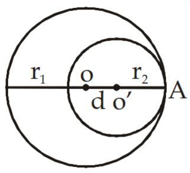
On adding (i) and (ii)

AD + AE = AB + BC + CA

2AD = AB + BC + CA

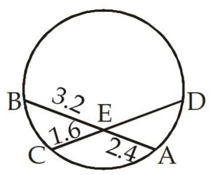
4. **(c);** Let O and O' be the centres of the two circles of radii r_1 and r_2 touching internally at A such that OO' = d.

Then OA – O'A = OO' $\Rightarrow$ $r_1 - r_2 = d$



5. **(a);**

6. **(c);**



AE × BE = CE × DE, 3.2 × 2.4 = 1.6 × DE

$$DE = \frac{3.2 \times 2.4}{1.6} = 4.8 \text{ cm}$$

7. **(c);** $\pi r^2 = 200$

$$r^2 = \frac{200}{\pi}, \quad r = \sqrt{\frac{200}{\pi}}, \quad R = 2r = 2\sqrt{\frac{200}{\pi}}$$

Area of langer circle = πR^2

$$= \pi \times 4 \times \frac{200}{\pi} = 800 \text{ cm}^2$$

8. **(b);** Diameter of circle = 10 cm

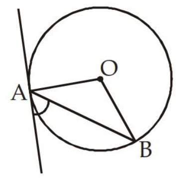
Diagonal of square = 10 cm

$$\text{Side} = \frac{\text{diagonal}}{\sqrt{2}} = \frac{10}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{10\sqrt{2}}{2} = 5\sqrt{2} \text{ cm}$$

9. **(b);** The focus of the tangent is clearly the equation of the tangent, which is $xx_1 + yy_1 = a^2$.

10. **(a);** Chords equidistant from the centre of a circle are equal.

11. **(c);**



$\angle OAB = 90° - \theta$

OA = OB

$\angle OAB = \angle ABO = 90° - \theta$

12. **(c);** Any cyclic parallelogram having equal adjacent sides is necessarily a rhombus.

13. **(a);** If a regular hexagon is inscribed in a circle of radius r, then its perimeter is 6r.

14. **(b);** Radius of circle = $\frac{1}{2} \times$ Diagonal of square

$$= \frac{1}{2} \times 4\sqrt{2} = 2\sqrt{2} \text{ cm}$$

15. **(a);** Let the length of chord AB is x

then $PQ^2 = AQ \cdot BQ$

$12^2 = (x + BQ) \times BQ$

$144 = (x + 8)8$

$x + 8 = 18, \quad x = 10$ cm

16. **(c);** $\angle BAD = \angle BCD$ (Angle made by same arc BC)

$\angle BDC = 38°$

OD = OC = r (radius of circle)

$\angle BDC = \angle OCD$

$\angle OCD = 38°$

17. **(b);** OB = OC $\Rightarrow$ $\angle OBC = \angle OCB = 20°$

$\angle BOC = 180° - 40° = 140°$

$$\angle BAC = \frac{1}{2}\angle BOC, \qquad \angle BAC = \frac{1}{2} \times 140°$$

$\angle BAC = 70°$

18. (d); $\angle ACB = \angle ADB = 20°$

$\angle ACB = \angle 20°$

$\angle CAB + \angle ACB + \angle ABC = 180°$

$85° + 20° + x = 180°$

$x = 180° - 105° = 75°$

19. (c); $\angle APB = \frac{1}{2}\angle AOB$

$\angle APB = \frac{1}{2} \times 90° = 45°$

20. (d); $AP \times CP = PD \times PB$

$14 \times 9 = 7 \times (7 + x)$

$18 = 7 + x, \qquad x = 11$ cm

21. (a); $x = 40°$ (Angle made by same arc AB.)

22. (b); $PT^2 = PA \cdot PB$

$12^2 = x \cdot (x + 7)$

$144 = x^2 + 7x$

$x^2 + 7x - 144 = 0$

$x^2 + 16x - 9x - 144 = 0$

$x(x + 16) - 9(x + 16) = 0$

$(x - 9)(x + 16) = 0$

$x = 9$ cm

23. (d); $PA = PB$

$\therefore \angle PAB = \angle PBA$

Also, $\angle PAB + \angle PBA + \angle APB = 180°$

$2\angle PAB = 120°$

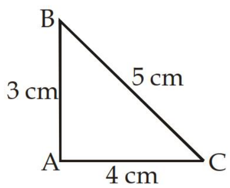
$\angle PAB = 60°$

Therefore ΔPAB is an equilateral triangle.

$AB = AP = 6$ cm

$AB = 6$ cm

24. (a);



Area of $\Delta ABC = \frac{1}{2} \times 3 \times 4 = 6$

Semiperimeter $= \frac{3+5+4}{2} = 6$

Inradius $= \frac{\text{Area}}{\text{Semiperimeter}} = \frac{6}{6} = 1$ cm

25. (b); $\angle AOB = \angle COD = 70°$ (Vertically opposite angle)

$\angle COD + \angle OCD + \angle ODC = 180°$

$2\angle OCD = 180° - 70°$

$2\angle OCD = 110°$

$\angle OCD = \frac{110°}{2} = 55°$

26. (c);

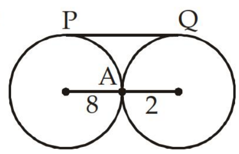
27. (c); $\angle CBA = \frac{1}{2}\angle COA = \frac{1}{2} \times 130° = 65°$

$\angle CBE = 180° - 65° = 115°$

28. (b); $\angle AOB + \angle APB = 180°$

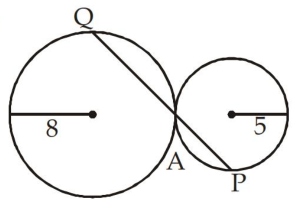
$\angle APB = \frac{1}{6} \times 180° = 30°$

29. (d);



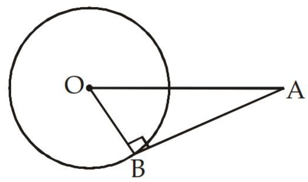
$PQ = 2\sqrt{r_1 r_2} = 2\sqrt{8 \times 2} = 2 \times 4 \text{ cm} = 8$ cm

30. (b);



Then the ratio $\frac{AP}{AQ} = \frac{5}{8} = 5:8$

31. (a);

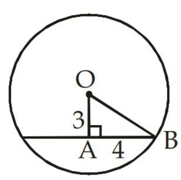


length of tangent

$AB = \sqrt{OA^2 - OB^2} = \sqrt{10^2 - 6^2}$

$= \sqrt{100 - 36} = \sqrt{64} = 8$ cm

32. (b);



In ΔOAB

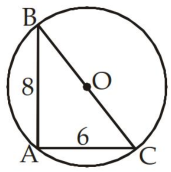
$OB^2 = OA^2 + AB^2$

$OB = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$ cm

33. (b); $OB^2 = \sqrt{OA^2 - AB^2}$

$= \sqrt{10^2 - 8^2} = \sqrt{100 - 64} = \sqrt{36} = 6$

34. (d);



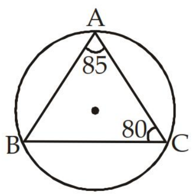
$\angle BAC = 90°$

BC is the diameter of the circle

$BC = \sqrt{AB^2 + AC^2} = \sqrt{8^2 + 6^2} = 10$ cm

Radius of circle = 5 cm

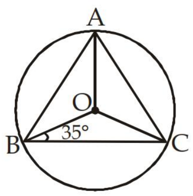
35. (b);



$\angle ABC = 180° - 80° - 85° = 15°$

$\Rightarrow \angle AOC = 2\angle ABC = 2 \times 15° = 30°$

36. (a);



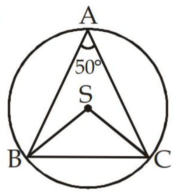
OB = OC = radius

$\angle OBC = \angle OCB = 35°$

$\angle BOC = 180° - 70° = 110°$

$\angle BAC = \frac{1}{2}\angle BOC = \frac{1}{2} \times 110 = 55°$

37. (b);



$\angle BAC = 50°, \quad \angle BSC = 100°$

BS = SC = radius

$\angle BCS = \frac{1}{2}(180 - 100) = 40°$

38. (d); Transverse Common tangent

$= \sqrt{(\text{Distance between centres}) - (r_1 + r_2)^2}$

$= \sqrt{10^2 - (4+4)^2} = \sqrt{10^2 - 8^2} = \sqrt{36} = 6$ cm

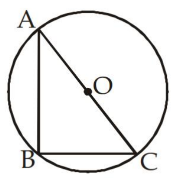
39. (d); Only one circle can pass through the non-collinear points.

40. (b); $3^2 + 4^2 = 5^2$

Therefore, ΔABC is a right angled triangle $\angle B$ = 90° = angle at the circumference

Diameter of circle = 5 cm

$\text{radius} = \frac{5}{2} = 2.5$ cm



Moderate

1. (c); $\angle KLN = 30°$ (Given)

$\angle LKN = 90°$ (angle made in semicircle)

$\angle KNL = 180° - 30° - 90°$

$\angle KNL = 60°$

$\angle KNL = \angle PKL$ (angle in alternate segment)

$\angle PKL = 60°$

2. (a); $\angle ADC = 120°$

$\angle ABC = 180° - 120°$

$\angle ABC = 60°$

In ΔABC

$\angle ABC = 60°, \angle ACB = 90°$

(angle is semicircle)

$\angle BAC = 180° - 60° - 90° = 30°$

3. (d); Let OM = x

In ΔOMB

$x^2 + 5^2 = r^2$ ... (i)

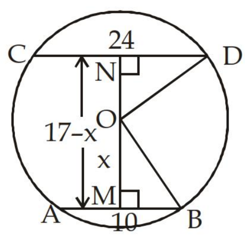
In ΔOND

$(17 - x)^2 + 12^2 = r^2$

$x^2 + 5^2 = (17 - x)^2 + 12^2$

$34x = 408$

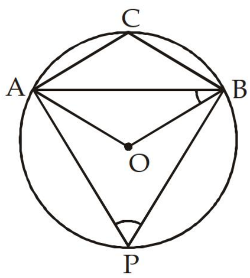
$\Rightarrow \quad x = \frac{408}{34} = 12$ cm



From (i)

$12^2 + 5^2 = r^2, \qquad r = \sqrt{169} = 13$ cm

4. **(c);** OA = OB = radius



$\angle OAB = \angle OBA = 25°$

$\angle AOB = 180° - 25° - 25° = 130°$

$\angle APB = \dfrac{130°}{2} = 65°$, $\angle ACB = 180° - 65° = 115°$

5. **(b);** $\angle PQO = 90°$, $\angle POR = 120°$

$\angle POQ = 180° - 120° = 60°$

In ΔQPO

$\angle OQP + \angle QOP + \angle QPO = 180°$

$\angle QPO = 180° - 90° - 60° = 30°$

6. **(d);** OB = OC = OA = radius

$\angle OBA = \angle OAB$, $\angle OAB = 20°$

Similarly $\angle OAC = 30°$

$\angle BAC = 20° + 30° = 50°$

$\angle BOC = 2 \times 50° = 100°$

7. **(c);** $\angle ABC = 80°$

$\angle ADC = 180° - 80° = 100°$

$\angle DAB = 60°$

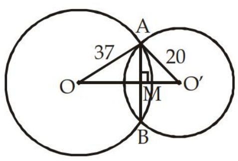
In ΔADQ

$\angle ADQ + \angle DAQ + \angle AQD = 180°$

$\angle AQD = 180° - 100° - 60°$

$\angle AQD = 20°$, $\angle BQC = 20°$

8. **(b);**



AM = 12 cm

In $\Delta AMO'$, $O'M^2 = O'A^2 - AM^2$

$O'M^2 = 20^2 - 12^2$

$O'M = 16$

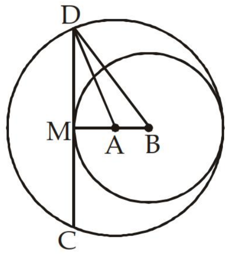
In AOM $OM^2 = 37^2 - 12^2$

$OM = 35$

then, $OO' = OM + MO'$

$= 35 + 16 = 51$ cm

9. **(a);** In ΔDAM



$AD^2 = AM^2 + DM^2$

$6^2 = 2^2 + DM^2$

$DM^2 = 6^2 - 2^2$

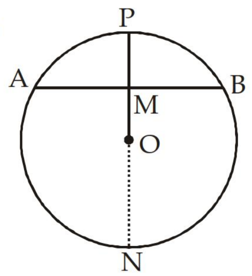
$DM^2 = 36 - 4$

$DM = \sqrt{32} = 4\sqrt{2}$

$CD = 2 \times DM$

$= 2 \times 4\sqrt{2} = 8\sqrt{2}$ cm

10. **(b);**



AB = 8 cm, MP = 2 cm

AM = 4 = MB, AM × MB = PM × MN

4 × 4 = 2 × MN, MN = 8 cm

MN = 2r – 2 = 8, 2r = 10, r = 5 cm

11. **(a);** $\angle ADC + \angle ABC = 180°$

$\angle ABC = 180° - 130° = 50°$

In ΔABC

$\angle ACB = 90°$, $\angle ABC = 50°$

$\angle CAB = 180° - 140° = 40°$

12. **(a);** $\angle POQ = 2 \times 50° = 100°$

$\angle PTQ = 180° - 100° = 80°$

13. **(c);** AB = AC and AD = CD

(distance midpoint)

$AD^2 = AP \times AB$

$\left(\dfrac{AB}{2}\right)^2 = AP \times AB$

$AB^2 = 4AP \times AB$

$\Rightarrow AB = 4AP$

14. **(a);** In ΔAOM and BOM

OM = (common), OA = OB (radius)

$\angle AOM = \angle BOM$, $\Delta AOM \cong \Delta BOM$

AM = BM,

$\dfrac{AM}{BM} = \dfrac{1}{1}$

15. **(a);** TP = TR and TQ = TP

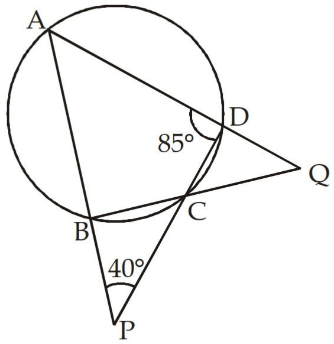
TQ = TP = TR

TQ : TR = 1 : 1

16. (b); $\angle AOC = 2 \times 60° = 120°$

$$\angle ABC = \frac{1}{2} \times \angle AOC = \frac{1}{2} \times 120° = 60°$$

17. (a);



$\angle ABC = 180° - 85° = 95°$

In ΔADP

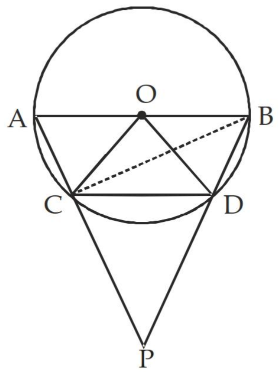
$\angle PAD = 180° - 85° - 40° = 180° - 125°$

$\angle PAD = 55°$

In ΔABQ

$\angle AQB = 180° - 55° - 95° = 30°$

18. (d);



$OC = OD = CD =$ radius

$\angle COD = 60°$

$$\angle CBD = \frac{1}{2} \times 60° = 30°$$

$\angle ACB = 90°$ (angle is semicircle)

$\angle APB = 180° - \angle BCP - \angle CBD$

$= 180° - 90° - 30°$

$\angle APB = 60°$

19. (b); $\angle ACB = 90°$

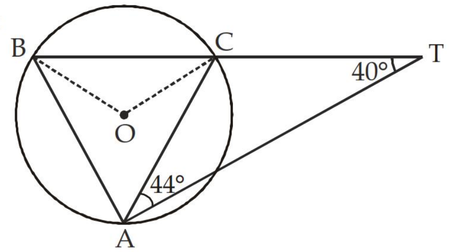
$$\angle CBD = \frac{1}{2} \times 31 = 15.5$$

$\angle APB = 180° - \angle PCB - \angle CBP$

$= 180° - 90° - 15.5°$

$\angle APB = 74.5°$

20. (d);



$\angle CAT = 44°$ (Given)

$\angle BTA = 40°$ (Given)

$\angle ACT = 180° - 44° - 40° = 96°$

$\angle CAT = \angle CBA = 44°$ (Alternate segment)

$\angle BCA = 180° - 96° = 84°$

$\angle BAC = 180° - 84° - 44 = 52°$

Angle subtended by BC at centre

$= 2 \times 52° = 104°$

21. (c); $\angle BOC = 2\angle BAC$

$OB = OC$

$\angle OBC = \angle OCB$

$\angle OBC = 180° - \angle BOC - \angle OCB$

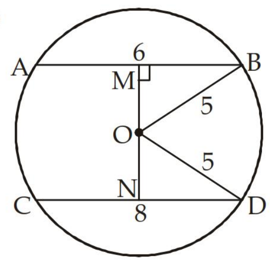
$= 180° - \angle BOC - \angle OBC$ $(\angle OBC = \angle OCB)$

$$= 90° - \frac{\angle BOC}{2}$$

$$\angle BAC + \angle OBC = \angle BAC + 90° - \frac{\angle BOC}{2}$$

$$= \frac{\angle BOC}{2} + 90° - \frac{\angle BOC}{2} = 90°$$

22. (b);



$$BM = \frac{1}{2} AB$$

$BM = 3$

$OM^2 = OB^2 - BM^2$

$OM^2 = 5^2 - 3^2 = 25 - 9 = 16$

$OM = 4$

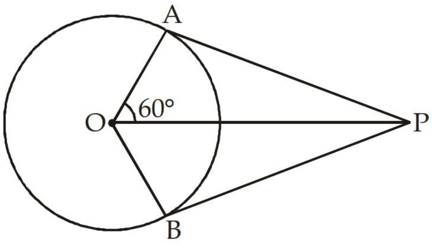
$ON^2 = OD^2 - DN^2$

$ON^2 = 5^2 - 4^2 = 25 - 16 = 9$

$ON = 3$

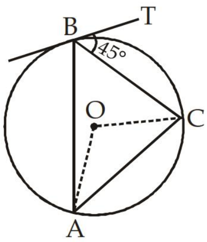
$MN = OM + ON = 3 + 4 = 7$ cm

23. (c);



$\angle AOP = 60°$ (Given)
$\angle OAP = 90°$ ($AP \perp OA$)
$\angle APO = 180° - 90° - 60°$
$\angle APO = 30°$
$\angle APB = 2\angle APO = 2 \times 30° = 60°$

24. (c);

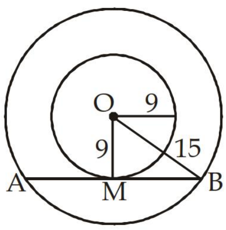


Let r = radius
CB = chord = 6 cm
$\angle CBT = 45°$ (Given)
$\angle CBT = \angle CAB$ (angle in a alternate segment)
$\angle COB = 2 \times 45° = 90°$
OC = OB = radius
In ΔCOB, $OC^2 + OB^2 = BC^2$
$2OC^2 = BC^2$

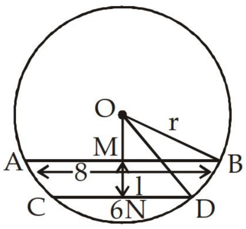
$$OC^2 = \frac{36}{2}$$

$r = OC = 3\sqrt{2}$ cm

25. (a); In ΔOBM
$OM^2 + BM^2 = OB^2$
$BM^2 = 15^2 - 9^2$
$BM^2 = 225 - 81$
$BM = \sqrt{144} = 12$ cm
$AB = 2\,BM$
$= 2 \times 12 = 24$ cm

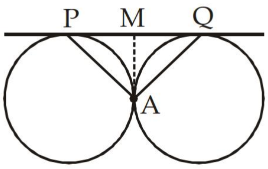


26. (a); Let OM = x
In ΔOMB
$r^2 = x^2 + 4^2$
In ΔOND
$r^2 = (x + 1)^2 + 3^2$
$x^2 + 4^2 = (x + 1)^2 + 3^2$
$x = 3$
$r^2 = 3^2 + 4^2 = 25$
$r = 5$ cm



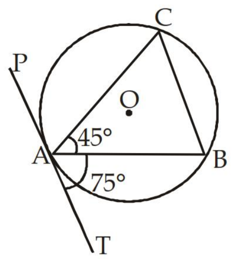
27. (d); Cannot be determined because value of Inradius and circumradius are side dependent unlike equilateral triangle.

28. (b);



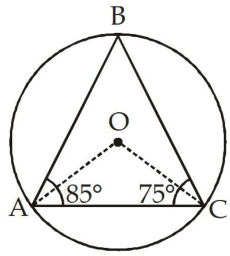
PM = AM
$\angle MPA = \angle MAP = x$
AM = MQ
$\angle MAQ = MQA = y$
In ΔAPQ
$\angle APQ + \angle AQP + \angle PAQ = 180°$
$x + y + \angle MAP + \angle MAQ = 180°$
$x + y + x + y = 180°$
$2(x + y) = 180°$
$x + y = 90°$
$\angle PAQ = 90°$

29. (c);



$\angle BAT = \angle ACB$
(angle in alternate segment)
$\angle ACB = 75°$
$\angle ABC = 180° - 45° - 75°$
$\angle ABC = 180° - 120° = 60°$

30. (c);

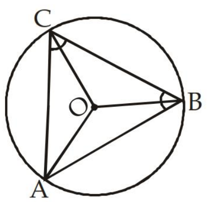


$\angle BAC = 85°$, $\angle BCA = 75°$
$\angle ABC = 180° - 85° - 75° = 180° - 160°$
$\angle ABC = 20°$, $\angle AOC = 40°$
$2\angle OAC = 180° - 40°$

$$\angle OAC = \frac{140°}{2} = 70°$$

Difficult

1. **(b);** In Δ OMB



Let AB = AC = a

BC = b

Area of $\Delta ABC = \frac{abc}{4R}$ where R is circumradius

for isosceles triangle = $\frac{a^2b}{4R}$

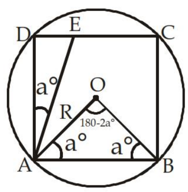
Area of triangle ABC = $\frac{1}{2} \times h \times BC$ where h is the height of isosceles triangle on BC

$$h = \frac{\sqrt{4a^2 - b^2}}{2}$$

$$\frac{6 \times 6 \times b}{4R} = \frac{1}{2} \times b \times \frac{\sqrt{4a^2 - b^2}}{2}$$

$$\frac{1296}{25} = 4a^2 - b^2, \quad b^2 = \frac{2304}{25}, \quad b = 9.6 \text{ cm}$$

2. **(c);**



OA = OB = radius

$\angle OAB = \angle OBA = a°$

Area of triangle AOB = $\frac{1}{2} R^2 \text{Sin}(180 - 2a°)$

$= \frac{1}{2} R^2 \text{Sin } 2a°$

Area (square ABCD) = $4 \times \frac{1}{2} R^2 \text{Sin } 2a°$

$= 2R^2 \text{ Sin } 2a$

$\frac{\text{Area(circle)}}{\text{Area(square)}} = \frac{\pi}{\sqrt{3}}$ (Given)

$$\frac{\pi R^2}{2R^2 \text{ Sin } 2a} = \frac{\pi}{\sqrt{3}}$$

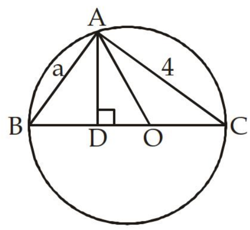
Sin 2a = Sin 60°

a = 30°

In ΔAED, $\text{Cos } a = \frac{AD}{AE}$

$$\frac{AE}{AD} = \frac{2}{\sqrt{3}}$$

3. **(b);** Let OD = x



In ΔABC

$AB^2 + AC^2 = BC^2$

$a^2 + 4^2 = BC^2$

$BC = \sqrt{a^2 + 16}$

$OB = \frac{1}{2} \times BC = \frac{1}{2}\sqrt{a^2 + 16}$

In ΔABC and ΔDBA

$$\frac{AB}{BC} = \frac{DB}{AB}$$

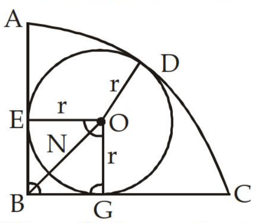
BC. DB = a^2

$$DB = \frac{a^2}{BC} = \frac{a^2}{\sqrt{a^2 + 16}}$$

$$OD = OB - DB = \frac{1}{2}\sqrt{a^2 + 16} - \frac{a^2}{\sqrt{a^2 + 16}}$$

$$= \frac{1}{2}\left[\frac{a^2 + 16 - 2a^2}{\sqrt{a^2 + 16}}\right] = \frac{16 - a^2}{2\sqrt{a^2 + 16}}$$

4. **(a);** Let radius of smaller circle = r



BE and BG are the tangent to smaller circle.

$OE \perp AB$ and $OG \perp BC$

$\angle B = \angle G = \angle E = 90°$

$\angle O = 90°$

Therefore OEBG is a square.

In ΔOGB

$OB^2 = OG^2 + BG^2 = r^2 + r^2$

$OB = \sqrt{2}\, r$

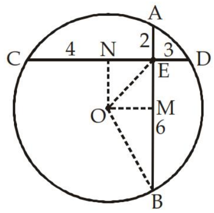
$OD = r$

$OB + OD = \sqrt{2}\, r + r$

$1 = r(\sqrt{2}+1)$

$r = \frac{1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1} = (\sqrt{2}-1)$ cm

5. (a);



$CE \times ED = AE \times EB$

$CE \times 3 = 2 \times 6$

$CE = 4$

$CN = \frac{1}{2} CD = \frac{1}{2}(4+3) = 3.5$

$NE = CD - ED - CN$

$= 7 - 3 - 3.5 = \cdot 5$

$AM = \frac{1}{2} AB = \frac{1}{2}(2+6) = 4$

$ME = AB - AE - MB$

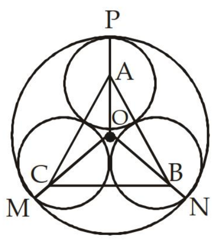
$= 8 - 2 - 4 = 2$

$OB^2 = OM^2 + MB^2$

$= \left(\frac{1}{2}\right)^2 + (4)^2 = \frac{1}{4} + 16 = \frac{65}{4}$

$OB = \frac{\sqrt{65}}{2} =$ radius

6. (c); Here $AB = BC = CA = 2$ units radius of each circle be 1 units



$OA = \frac{2}{3} \times AB \times \frac{\sqrt{3}}{2} = \frac{2}{3} \times \frac{\sqrt{3}}{2} \times 2 = \frac{2}{\sqrt{3}}$ cm

$OP = OA + AP = \frac{2}{\sqrt{3}} + 1 = \frac{2+\sqrt{3}}{\sqrt{3}}$ cm

Area of circumscribing circle $= \pi r^2$

$\frac{\pi}{3}(2+\sqrt{3})^2$

7. (b); $\angle DBC = 33°$

$\angle DBC = x = 33°$

(alternate segment theorem)

$z = 180° - 33° = 147°$

$BE = BD$

$\angle BED = \angle BDE$

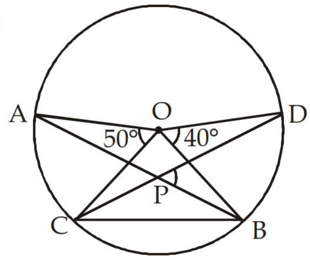
$\angle BDE = 33°$

$\angle ABE = 33°$ (alternate segment theorem)

$y = 90° - 33° = 57°$

$x + y + z = 33° + 57° + 147° = 237°$

8. (c);



First join BC

$\angle BCP = \frac{1}{2} \times \angle BOD = \frac{1}{2} \times 40°$

$\angle BCP = 20°$

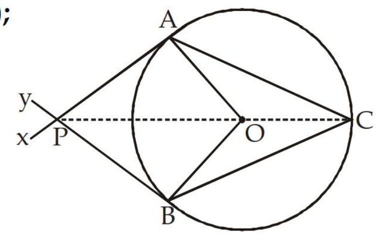
$\angle ABC = \frac{1}{2} \times \angle AOC = \frac{1}{2} \times 50° = 25°$

$\angle BPD = \angle BCP + \angle CBP$

(Exterior angle to triangle PBC)

$= 20° + 25° = 45°$

9. (a);



$\angle ACB = 65°$

$\angle AOB = 2\times 65° = 130°$

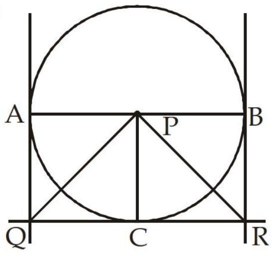
$\angle POA = \frac{1}{2} \times 130° = 65°$

$\angle POA = 65°$

$\angle PAO = 90°$

$\angle APO = 180° - 65° - 90° = 25°$

10. (c);



In ΔPCR and ΔRBP

PC = PB radius

RC = RB

PR is common

$\Delta PCR \cong \Delta PBR$

$\angle CPR = \angle BPR$

Similarly

$\Delta CPQ \cong \Delta APQ$, $\angle APQ = \angle CPQ$

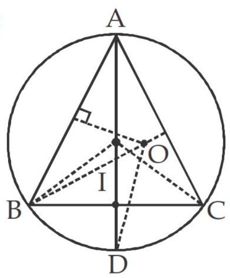
$\angle APQ + \angle CPQ + \angle CPR + \angle BPR = 180°$

$2\angle CPQ + 2\angle CPR = 180°$

$\angle CPQ + \angle CPR = 90°$

$\angle QPR = 90°$

11. (a);



$\angle BOD = Z$

$\angle BAD = \frac{1}{2}Z$

$\angle ABI = \angle IBC = \frac{x}{2}$ [$\therefore$ I is the incentre]

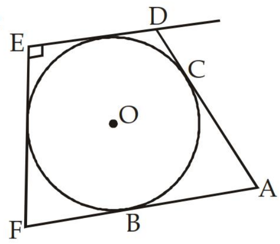
Now in ΔABI

$y = \angle BAI + \angle ABI = \frac{z}{2} + \frac{x}{2} = \frac{x+z}{2}$

$x + z = 2y$

$\frac{x+z}{3y} = \frac{2y}{3y} = \frac{2}{3}$

12. (c);



Here $OM \perp DE$

and $ON \perp EF$

AB = 27,

$\therefore$ AC = AB = 27

(tangent drawn from a external point are equal)

AD = 38

CD = 38 – 27 = 11 cm

DE = 24

EM = 24 – 11 = 13 cm

In ☐ EMON

EM = EN, $\angle E = 90°$, $\angle EMO = 90°$

Therefore it is a square

ON = EM = 13 cm

radius of Circle is 13 cm

13. (d); $\Delta PQS \sim \Delta PRT$

$\frac{PQ}{PR} = \frac{QS}{RT}$, $\frac{24}{PR} = \frac{7}{21}$

$PR = \frac{24 \times 21}{7} = 72 \text{ cm}$

In ΔPQS

$PS^2 = PQ^2 + QS^2$

$PS^2 = 24^2 + 7^2$, PS = 25

$PT^2 = PR^2 + RT^2$

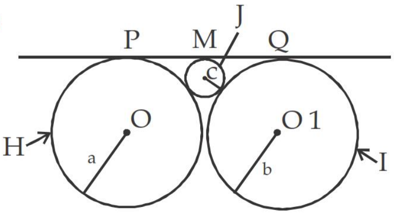
$PT^2 = 72^2 + 21^2$

PT = 75

ST = 75 – 25 = 50 cm

$SN = \frac{1}{2} \times ST = \frac{1}{2} \times 50 = 25 \text{cm}$

14. (d);



PQ is the common tangent of two bigger circle H and I.

Therefore PQ = $2\sqrt{ab}$

Similarly for circle H and I

PM = $2\sqrt{ac}$

Similarly, MQ = $2\sqrt{bc}$

PQ = PM + OM

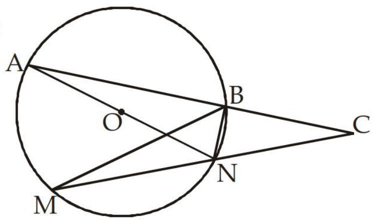
$2\sqrt{ab} = 2\sqrt{ac} + 2\sqrt{bc}$

$\sqrt{ab} = \sqrt{ac} + \sqrt{bc}$

On dividing both sides by $\sqrt{abc}$

$$\frac{1}{\sqrt{c}} = \frac{1}{\sqrt{b}} + \frac{1}{\sqrt{a}}$$

15. (d);



$\angle BAN = 38°$

$\angle BMN = 38°$ (Angle b/w same arc)

$\angle ABN = 90°$

In ΔBNC

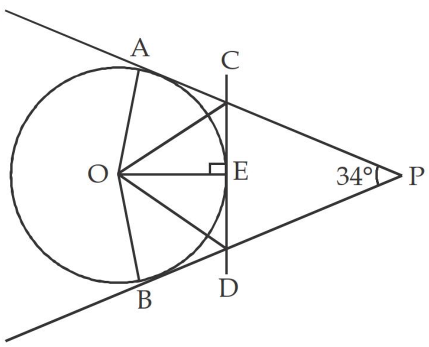
$\angle BNC = 90° - 20° = 70°$

$\angle BNM = 180° - 70° = 110°$

In ΔMBN

$\angle MBN = 180° - 110° - 38° = 32°$

16. (c);



$\angle AOB = 180° - 34° = 146°$

In ΔOAC and ΔOEC

OA = OE = radius

and CA = CE

(tangent from external point)

$\Delta OAC \cong \Delta OEC$

$\angle AOC = \angle COE$

Similarly

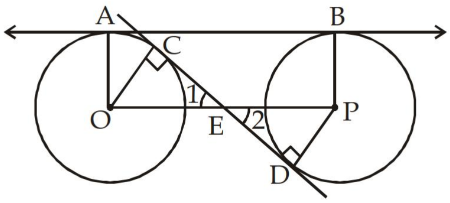
$\angle BOD = \angle DOE$

$2\angle COE + 2\angle DOE = 146°$

$2(\angle COE + \angle DOE) = 146°$

$\angle COE + \angle DOE = 73° = \angle COD = 73°$

17. (c);



CD = 24 cm

OC = 5

Circles have same radius

So, CE = $\frac{1}{2}$CD and OE = $\frac{1}{2}$OP

In ΔOCE, $12^2 + 5^2 = OE^2$

$OE^2 = 13^2$, OE = 13

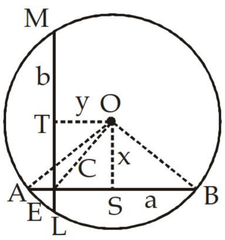
OP = 2 × 13 = 26 cm

OA = BP and $\angle OAB = \angle PBA = 90°$

Therefore AOPB is a rectangle.

OP = AB , OP = 26 cm

18. (c);



Let AB = 2a and ML = 2b

OS $\perp$ AB and OT $\perp$ ML

AS = BS = a

MT = TL = b

$\angle OSA = \angle OTE = \angle TES = 90°$

Therefore TESO is a rectangle.

OS = x and OT = y

In ΔOSB, $x^2 + a^2 = r^2$

In ΔOMT, $y^2 + b^2 = r^2$

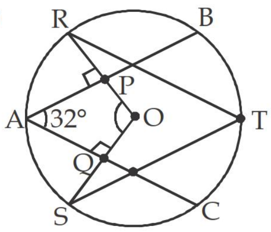
On adding above equation

$2r^2 = x^2 + y^2 + a^2 + b^2$

$\Rightarrow \quad 2r^2 = a^2 + b^2 + c^2$

$$\Rightarrow \quad r = \sqrt{\frac{a^2 + b^2 + c^2}{2}}$$

19. (b);



APOQ is a quadrilateral

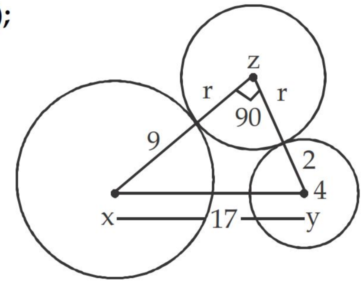
$\angle PAQ = 32°$

$\angle APO = \angle AQO = 90°$

$\angle POQ = 180° - 32°, \quad \angle POQ = 148°$

$$\angle RTS = \frac{148°}{2} = 74°$$

20. (d);



Here, $xz = 9 + r$

$xy = 17, \qquad yz = 2 + r$

In Δ xyz

$(xy)^2 = (xz)^2 + (yz)^2, \quad 17^2 = (9 + r)^2 + (2 + r)^2$

$289 = 81 + r^2 + 18r + 4 + r^2 + 4r$

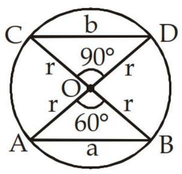
$2r^2 + 22r - 204 = 0, \quad r^2 + 11r - 102 = 0$

$r^2 + 17r - 6r - 102 = 0, \quad r(r + 17) - 6(r + 17) = 0$

$(r - 6)(r + 17) = 0, \quad r = 6$ cm

Previous Year (Memory Based)

1. (a);



In ΔAOB, $\angle AOB = 60°$

$\angle OBA = \angle OAB = 60°$

ΔAOB is an equilateral triangle

$r = a$ ... (i)

In ΔCOD, $\angle COD = 90°$

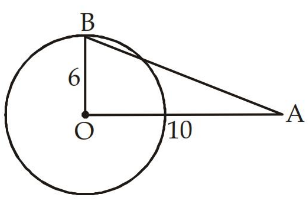
$b^2 = r^2 + r^2$

$$r = \frac{b}{\sqrt{2}} \quad \text{... (ii)}$$

On solving (i) and (ii)

$$\frac{b}{\sqrt{2}} = a, \quad b = \sqrt{2}a$$

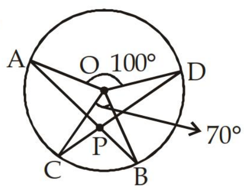
2. (d);



Length of tangent $AB = \sqrt{OA^2 - OB^2}$

$= \sqrt{10^2 - 6^2} = \sqrt{64} = 8$ cm

3. (d);



$\angle AOD = 100°, \quad \angle COB = 70°$

Join AC

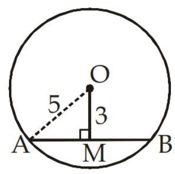
$$\angle CAB = \frac{1}{2}\angle COB = \frac{1}{2} \times 70° = 35°$$

$$\angle ACD = \frac{1}{2}\angle AOD = \frac{1}{2} \times 100° = 50°$$

$\angle APC = 180° - 35° - 50° = 95°$

4. (b); Distance between their centres $= 6.3 - 3.6 = 2.7$cm

5. (c); Here $OM \perp AB$



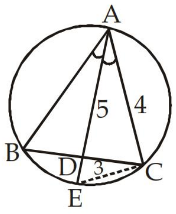
$AM = \sqrt{OA^2 - OM^2}$

$= \sqrt{25 - 9}$

$AM = \sqrt{16} = 4$

length of chord $= 2 \times AM = 2 \times 4 = 8$cm

6. (b);

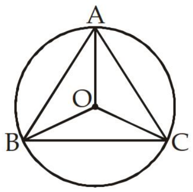


In ΔABD and ΔAEC

$$\frac{AB}{AE}=\frac{AD}{AC},\quad \frac{AB}{8}=\frac{5}{4}$$

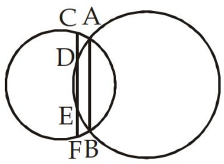
$$AB=\frac{8\times5}{4}=10\text{ cm}$$

7. (b);



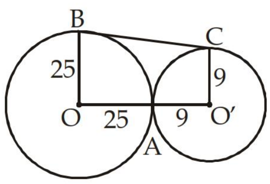
Since ΔABC is an equilateral triangle.
Therefore, angle formed at centre are equal
$\therefore\ \angle BOC=\angle AOC=\angle AOB=120°$

8. (c);



Given AB | | CF and CD = 4.5
CD = EF $\Rightarrow$ EF = 4.5 cm

9. (b);

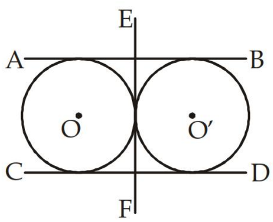


OO' = 25 + 9 = 34 cm

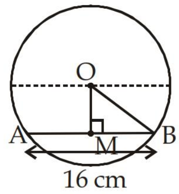
$$BC=\sqrt{(OO')^2-(OB-O'C)^2}$$

$$=\sqrt{34^2-16^2}=\sqrt{900}=30\text{ cm}$$

10. (c);



11. (a);



In ΔOBM,

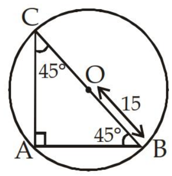
OB = radius of circle, $\quad OB=\frac{20}{2}=10\text{ cm}$

$$BM=\frac{1}{2}\times AB=8\text{ cm}$$

By using pythagorus theorem

$$OM=\sqrt{10^2-8^2}=6\text{ cm}$$

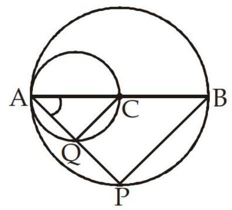
12. (c);



Given $\angle ABC=\angle ACB=45°$
Third angle $\angle CAB=180°-(45°+45°)=90°$
Since, circum radius = OB = OC = OA = 15 cm
AB = AC
By using pythagorus theorem
$AB^2+AC^2=BC^2$
$2AB^2=30^2$

$$AB=\frac{30}{\sqrt{2}}=\frac{30}{\sqrt{2}}\times\frac{\sqrt{2}}{\sqrt{2}}=15\sqrt{2}\text{cm}$$

13. (a);

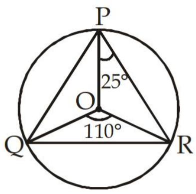


In ΔAQC and ΔAPB
$\angle AQC=\angle APB$ (angle made is semicircle)
$\angle QAC=\angle PAB$ (common)
$\Delta AQC\sim\Delta APB$

$$\frac{AQ}{AP}=\frac{AC}{AB}$$

QC | | PB

14. (c);



$\angle ORP = \angle OPR = 25°$

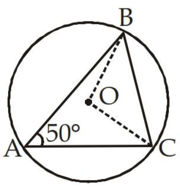
$\angle ORQ + \angle OQR + \angle QOR = 180°$

$2\angle ORQ = 180° - 110°$

$\angle ORQ = \dfrac{70°}{2} = 35°$

$\angle PRQ = \angle ORP + \angle ORQ = 25° + 35° = 60°$

15. (c);



$\angle BOC = 2\angle BAC$

$\angle BOC = 100°$

{angle subtended at the centre is twice the angle subtended at point A}

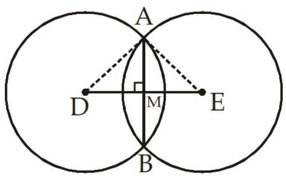
$\angle OBC = \angle OCB \qquad \{\therefore\ OB = OC = \text{radius}\}$

Now, In ΔBOC

$\angle OBC + \angle OCB + 100° = 180°$

$2\angle OBC = 80°, \quad \angle OBC = 40°$

16. (c);



Join D and E and AD and AE

In ΔADM By pythagorus theorem

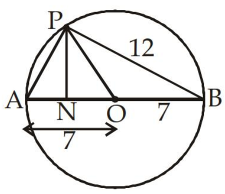
$AD^2 = AM^2 + DM^2 \qquad \left\{\therefore AM = \dfrac{1}{2}AB\right\}$

$5^2 = 4^2 + DM^2$

$DM = 3$

$DE = 2DM = 2 \times 3 = 6$ cm

17. (c);



In ΔAPB,

By using pythagorus theorem.

$AB^2 = AP^2 + PB^2, \quad 14^2 = AP^2 + 12^2$

$AP^2 = 196 - 144, \quad AP^2 = 52$

Now, let $AN = a$ cm

Then, $NO = (7 - a)$ cm

In ΔAPN, $AP^2 = a^2 + PN^2$

$PN^2 = 52 - a^2 \qquad$ ...(i)

In ΔPNO, $PO^2 = PN^2 + NO^2$

$7^2 = PN^2 + (7 - a)^2$

$49 - (7 - a)^2 = PN^2 \qquad$ ...(ii)

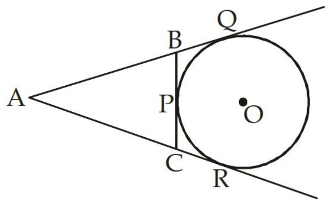
From equation (i) and (ii)

$14x = 52, \quad x = \dfrac{52}{14} = \dfrac{26}{7}$ cm

Length of NO $= 7 - \dfrac{26}{7} = \dfrac{49-26}{7} = \dfrac{23}{7}$ cm

Length of NB $= 7 + \dfrac{23}{7} = \dfrac{49+23}{7} = \dfrac{72}{7} = 10\dfrac{2}{7}$ cm

18. (c);



Here AQ = AR, BQ = BP and CR = CP

Perimeter of $\Delta ABC = AB + BC + AC$

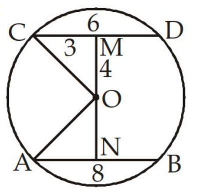
$= AB + BP + CP + AC$

$= AB + BQ + CR + AC$

$= AQ + AR$

$= 2AQ = 2 \times 6 = 12$ cm

19. (d);



Radius of circle

$OC^2 = MC^2 + OM^2$

$= 3^2 + 4^2 = 9 + 16 = 25$

$OC = \sqrt{25} = 5$ cm

In ΔONA, $OA^2 = AN^2 + ON^2$

$ON^2 = 5^2 - 4^2 = 25 - 16$

$ON = \sqrt{9} = 3$ cm

20. (b); Given, $\angle BOD = 15°$, $\angle EOA = 85°$

$\angle EOD = 180° - (85° + 15°) = 80°$

In ΔEOD, EO = OD (radius)

$\angle OED = \angle ODE$

$\angle OED = 50°$

Now In ΔOEC

$\angle OEC + \angle EOC + \angle OCE = 180°$

$50° + 95° + \angle OCE = 180°$

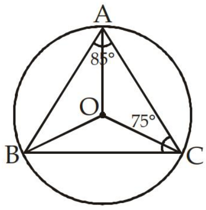
$\angle OCE = 180° - 50° - 95°$

$\angle OCE = 35°, \quad \angle ECA = 35°$

21. (b); In radius $= \dfrac{\text{Side}}{2\sqrt{3}}$

$3 = \dfrac{\text{Side}}{2\sqrt{3}}, \quad \text{Side} = 6\sqrt{3}\text{ cm}$

22. (c);



In ΔABC

$\angle A + \angle B + \angle C = 180°$

$\angle B = 180° - 85° - 75° = 20°$

$\angle AOC = 2\angle B = 40°$

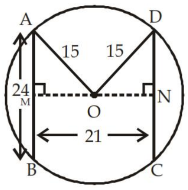
In ΔAOC, $\angle OAC + \angle OCA + \angle AOC = 180°$

$2\angle OAC = 180° - 40°$

$2\angle OAC = 140°$

$\angle OAC = 70°$

23. (b);



In ΔAOM, $\angle AMO = 90°$

$AM = 12, \quad OA = 15$

$OM = \sqrt{OA^2 - AM^2} = \sqrt{15^2 - 12^2}$

$OM = \sqrt{225 - 144} = \sqrt{81}$

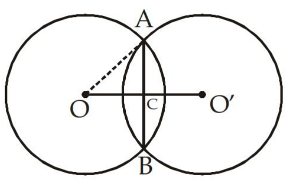
$OM = 9\text{ cm}$

$ON = 21 - 9 = 12\text{ cm}$

In ΔDON, $DN = \sqrt{15^2 - 12^2} = 9\text{ cm}$

Length of chord $CD = 2 \times DN = 2 \times 9 = 18\text{ cm}$

24. (a);



$OO' = 12\text{ cm}, \quad AB = 16\text{ cm}$

$AC = BC = 8\text{ cm}, \quad OC = O'C = 6\text{ cm}$

In ΔOAC

$OA = \sqrt{OC^2 + CA^2} = \sqrt{6^2 + 8^2}$

$= \sqrt{36 + 64} = \sqrt{100}$

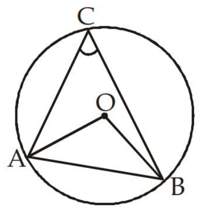
$OA = 10\text{ cm}$

25. (d); Since,

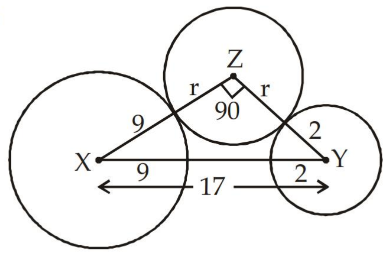
$AB = OB = OA = \text{radius}$

$\angle AOB = 60°$

$\angle ACB = \dfrac{1}{2} \times 60° = 30°$



26. (d);



In ΔXYZ, By using phythagorus theorem

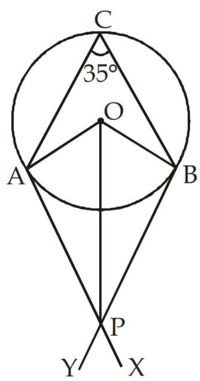
$XY^2 = XZ^2 + YZ^2$

$17^2 = (9 + r)^2 + (2 + r)^2$

$289 = 81 + r^2 + 18r + 4 + r^2 + 4r$

$r^2 + 11r - 102 = 0, \quad r = 6\text{ cm}$

27. (a);



$\angle OAP = 90°$

Angle between tangent and radius of circle

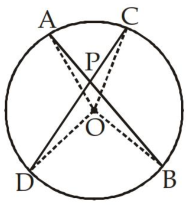
$\angle AOB = 2\angle ACB = 2 \times 35°$

$\angle AOB = 70°$

$\angle AOP = \dfrac{70°}{2} = 35°$

$\angle APO = 180° - 90° - 35° = 55°$

28. (c);



$\angle BOC = 2\angle BAC$ ...(i)

$\angle AOD = 2\angle DCA$ ...(ii)

From (i) and (ii)

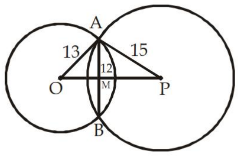
$\angle BOC + \angle AOD = 2(\angle BAC + \angle DCA) = 2\angle BPC$

$2\angle BPC = 20° + 30°$, $\angle BPC = 25°$

29. (c); Length of transverse tangent

$= \sqrt{10^2 - (3+3)^2} = \sqrt{100-36} = \sqrt{64} = 8$ cm

30. (b);



In ΔAOM

$OA^2 = AM^2 + OM^2$, $13^2 = 12^2 + OM^2$

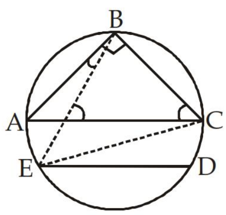
$OM = 5$ cm

In ΔAPM, $AP^2 = AM^2 + MP^2$, $MP^2 = 15^2 - 12^2$

$MP = 9$ cm

$OP = OM + MP = 9 + 5 = 14$ cm

31. (d);



$\angle CBE = 50°$ (given), $\angle ABE = 90° - 50°$

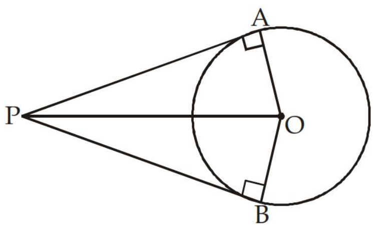
$\angle ABE = 40°$, $\angle ABE = \angle ACE$

(Angle made by same arc are equal)

$AC || ED$, $\angle ACE = \angle DEC = 40°$

32. (c); radius $= \sqrt{3^2 + 4^2} = 5$ cm

33. (c);



Let the radius of circle be r

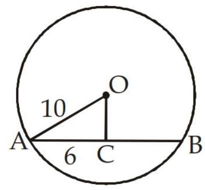
$OP = 2r$

In ΔAOP, $\sin\angle APO = \frac{OA}{OP} = \frac{r}{2r} = \frac{1}{2}$

$\sin\angle APO = \frac{1}{2}$, $\angle APO = 30°$

$\angle APB = 2 \times \angle APO = 2 \times 30° = 60°$

34. (d); Radius $= \frac{\text{Diameter}}{2} = \frac{20}{2} = 10$ cm



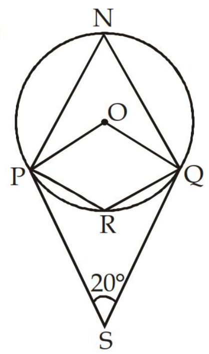
$OC^2 = OA^2 - AC^2$

$OC^2 = 10^2 - 6^2$

$OC^2 = 100 - 36$

$OC = \sqrt{64} = 8$ cm

35. (d);



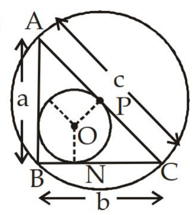
Given, $\angle PSQ = 20°$

$\angle POQ = 180° - 20° = 160°$

$\angle PNQ = \frac{1}{2} \times \angle POQ = \frac{1}{2} \times 160° = 80°$

$\angle PRQ = 180° - 80° = 100°$

36. (b);



$PC = 15$ cm = circum radius

$AC = 2 \times 15 = 30$ cm

$ON = 6$ = Inradius

Inradius $= \frac{a+b-c}{2}$

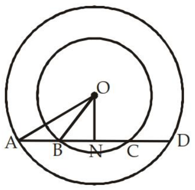
$a + b - c = 12$, $a + b = 12 + c$

$a + b = 42$ cm

Now, check by option

Here, $a = 18$ and $b = 24$, $c = 30$

37. (c);



According to question

Given, $BC = 12 \quad OA = 17$ cm, $\quad OB = 10$ cm

$BN = NC = 6$ cm

In right angle ΔONB.

$OB^2 = ON^2 + BN^2, \quad 10^2 = ON^2 + 6^2$

$ON^2 = 64, \quad ON = 8$ cm

In ΔONA, $OA^2 = ON^2 + AN^2$

$17^2 = 8^2 + AN^2, \qquad AN = 15$ cm

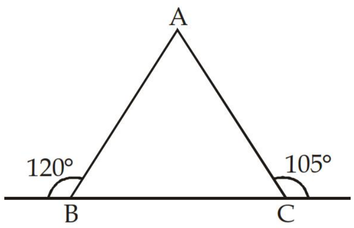
$AD = 2 \times AN = 2 \times 15 = 30$ cm

38. (a); $TCT = \sqrt{d^2 - (r_1 + r_2)^2}$

$(8)^2 = d^2 - 9^2$

$d^2 = 64 + 81, \quad d = \sqrt{145}$

39. (c);



$\angle ABC = 180° - 120°$

$\angle ABC = 60°$

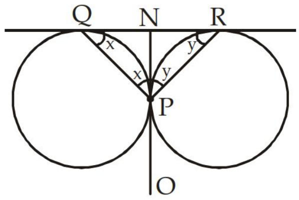
$\angle ACB = 180° - 105°$

$\angle ACB = 75°$

$\angle ABC + \angle ACB + \angle BAC = 180°$

$\angle BAC = 180° - 75° - 60° = 45°$

40. (c);



$QN = NP = NR$

In ΔQPN, $\qquad \angle NQP = \angle NPQ$

$\qquad \angle NRP = \angle NPR$

In ΔNPR, $\qquad \angle P + \angle Q + \angle R = 180°$

$x + y + x + y = 180°$

$2(x + y) = 180°$

$x + y = 90°$

$x + y = \angle P = 90°$